

VISOKA ŠKOLA STRUKOVNIH STUDIJA ZA INFORMACIONE I KOMUNIKACIONE TEHNOLOGIJE

DOKUMENTACIJA

WEB PROGRAMIRANJE 2

STUDENT:

MARKO PEŠIĆ 86/23

Sadržaj

1.1. Uvod - Tehnički aspekti

1.1.1. Tehnologije

1.1.2. Template sajta

1.1.3. Generisanje slika

1.2. Uvod - Opis sajta

1.1.4. Sitniji detalji:

1.3. Uvod – Struktura fajlova

2.1. Pregled sajta [Vizuelni pregled (uredjaji >768px (tableti i veći ekran)]

2.2. Pregled sajta [Vizuelni pregled (uredjaji < 768px)]

3.1. Ključne funkcije (opis)

4.1. Ceo kod.

4.1.1. HTML

4.1.2. JS

4.1.3. JSON

1.1. Uvod - Tehnički aspekti

1.1.1. Tehnologije

Jezici: HTML, CSS, JS

Biblioteka: Bootstrap, Node.js (Za generisanje PC komponenti)

Radno okruženje: Visual Studio Code

Host: GitHub

1.1.2. Template sajta: nije preuzet

1.1.3. Generisanje slika:

Za generisanje slika korišćenih za sajt, upotrebljena je node.js skripta koja gadjja *placeholder.co* sajt za privremene slike.

Skripta prolazi kroz components.json i generiše sivi (hex: 2a2a2a) kvadrat (400px * 400px) sa belim tekstom koji ima brend komponente (iz components.json) i ispod ime kategorije kojoj komponenta pripada (npr. AMD Ryzen (brend), pa ispod Processor (ime kategorije)).

Slike su onda sačuvane u assets/images da se ne bi zavisilo od dostupnosti sajta placeholder.co.

Svaka slika ima naziv koji odgovara id-u komponente pa .png (npr. cpu_1.png). (Cela skripta će biti u sekciji „Kod“)

1.1.4. Sitniji detalji:

Za ikonice su korišćeni Bootstrap i Font Awesome.

1.2. Uvod - Opis sajta

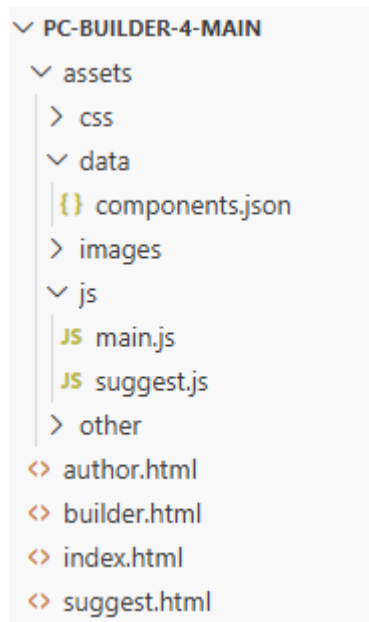
Sajt je PC Builder aplikacija / Konfigurator.

Sadrži 100 komponenti kroz 10 kategorija po 10 komponenti:

CPU, GPU, PSU, Matične ploče, RAM memorije, Skladišta, Miševi, Tastature, Slušalice, Monitori

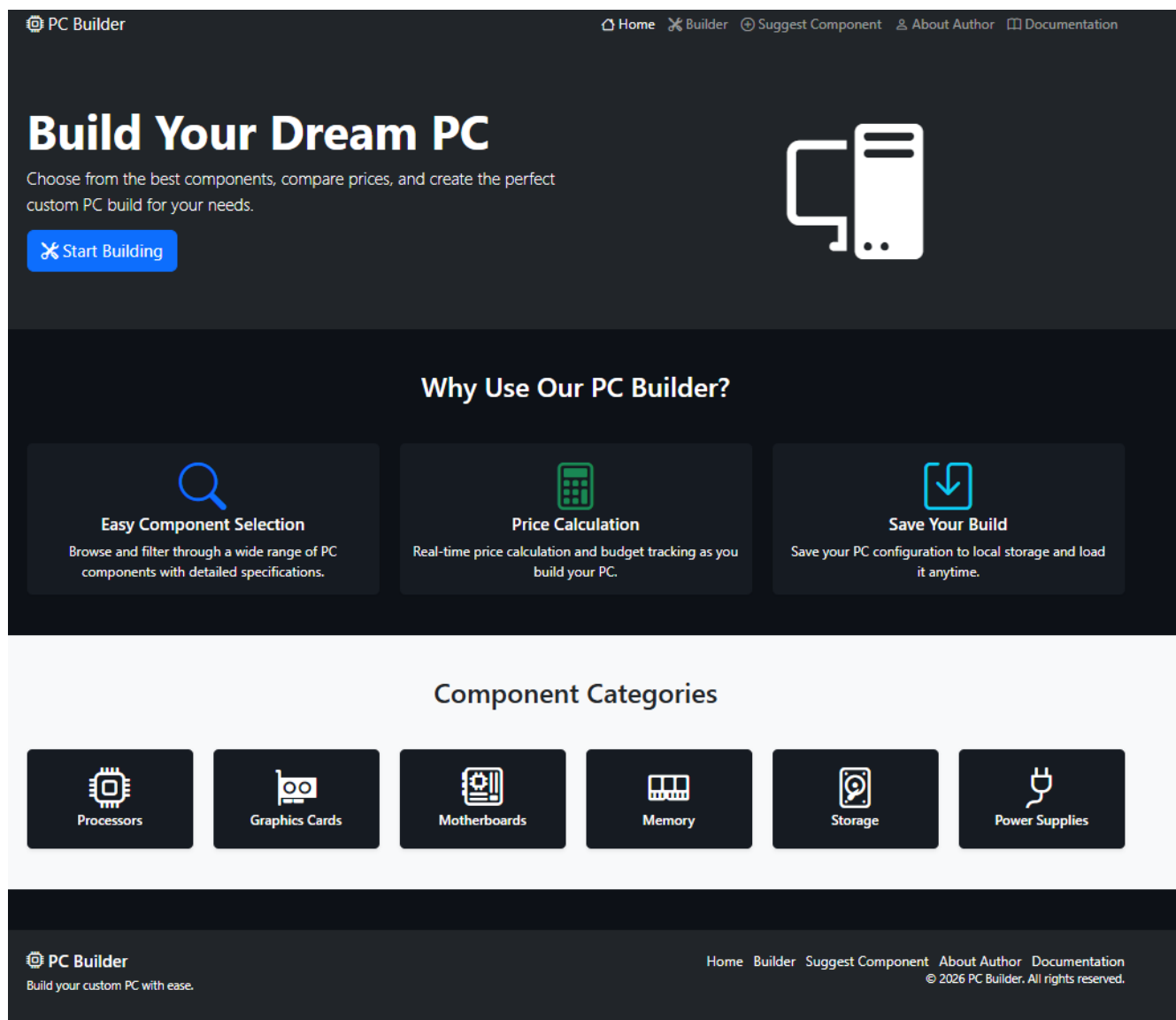
Glavna funkcija sajta je „Your Build“. Korisnici mogu da dodaju komponente u svoj build, sačuvaju build (lokalno), omiljene komponente (lokalno) i aplikacija na osnovu interne logike (obrađeno dalje u dokumentaciji) prikazuje kompatibilnost između komponenti.

1.3. Uvod – Struktura fajlova



Slika 1. File tree

2.1. Pregled sajta [Vizuelni pregled (uredjaji >768px (tableti i veći ekrani))]



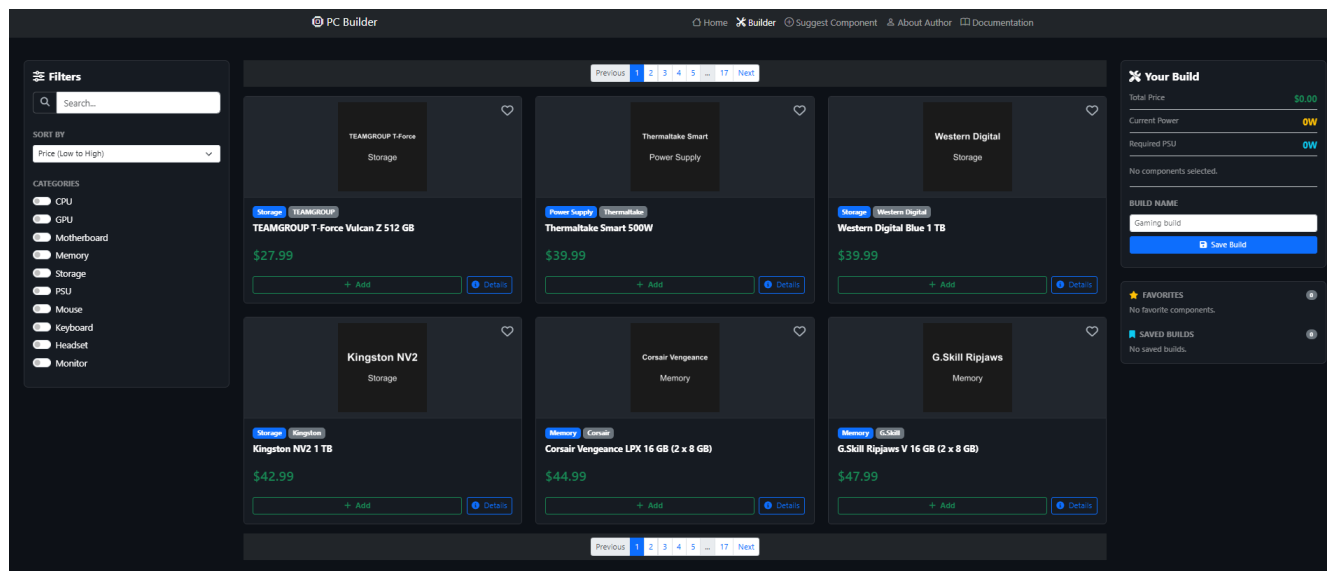
Slika 2. index.html

Početna strana (index.html):

Početna strana sadrži navigaciju, glavni deo i footer.

Navigacija ima linkove ka builder-u, suggest components, stranicu o autoru i link ka ovoj dokumentaciji.

Klikom na dugme *Start Building* odlazimo na builder.html. Ako korisnik klikne na jednu od kategorija, odlazi takodje odlazi na builder.html ali mu je kliknuta kategorija već izabrana.



Slika 3. builder.html

Glavna stranica sajta. Kao što je pomenuto, korisnik bira komponente klikom na *+Add* dugme. Klikom na dugmiće u *Categories*, prikazuju se samo komponente izabrane kategorije. Ako izabere više kategorije, sve komponente izabranih kategorija će biti prikazane. Klikom na *Sort By* padajuću listu prikazuju se opcije: Price (low to high), Price (high to low), Ime (A-Z), Ime (Z-A), Datuma kada je komponenta izašla na tržište (najstarije i najmladje). Svaka komponenta ima *Details* dugme koja otvara modal sa dodatnim informacijama. (Slika 4.)

Intel Core
Processor

Intel Core i5-12400F 2.5 GHz 6-Core Processor

\$149.99

Processor

Intel

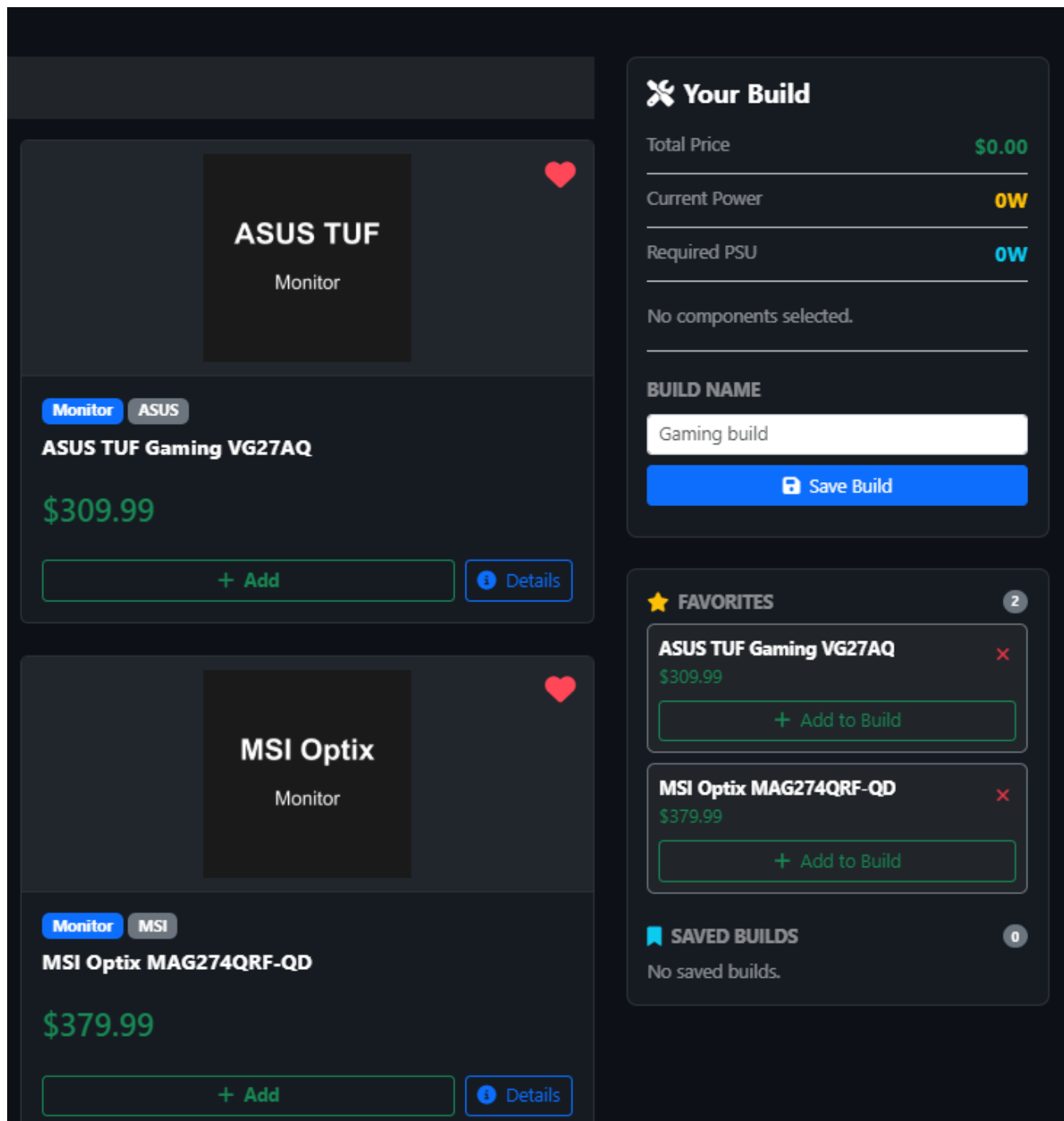
+ Add to Build

Specifications

Cores:	6
Threads:	12
Base Clock:	2.5 GHz
Boost Clock:	4.4 GHz
Tdp:	65W
Socket:	LGA1700
CACHE	
L1:	480 KB
L2:	7.5 MB
L3:	18 MB

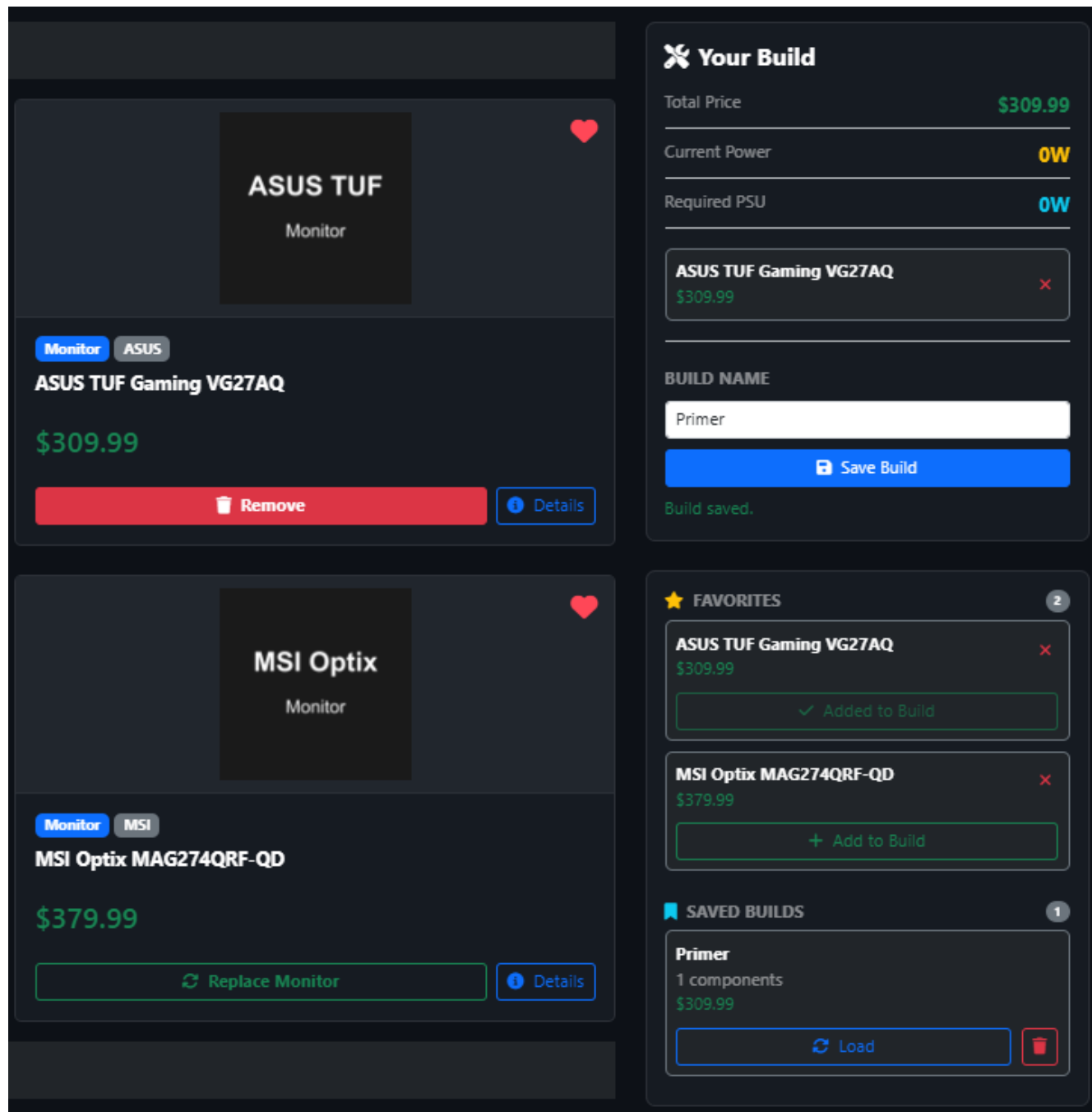
Slika 4. Modal Primer

Sobzirom da svaka komponenta ima objekte u svom datom objektu, koristio sam rekurzivnu funkciju koja se poziva ako je u objektu jos jedan objekat (npr. *Cache* (slika 4.)) je objekat.



Slika 5. Favorite components primer

Klikom na srce ikonicu, srce se popuni i komponenta se dodaje u *Favorites*, i postaje dostupna za dodavanje u build. Klikom na X, komponenta se briše iz *Favorites*. (Slika 5.)



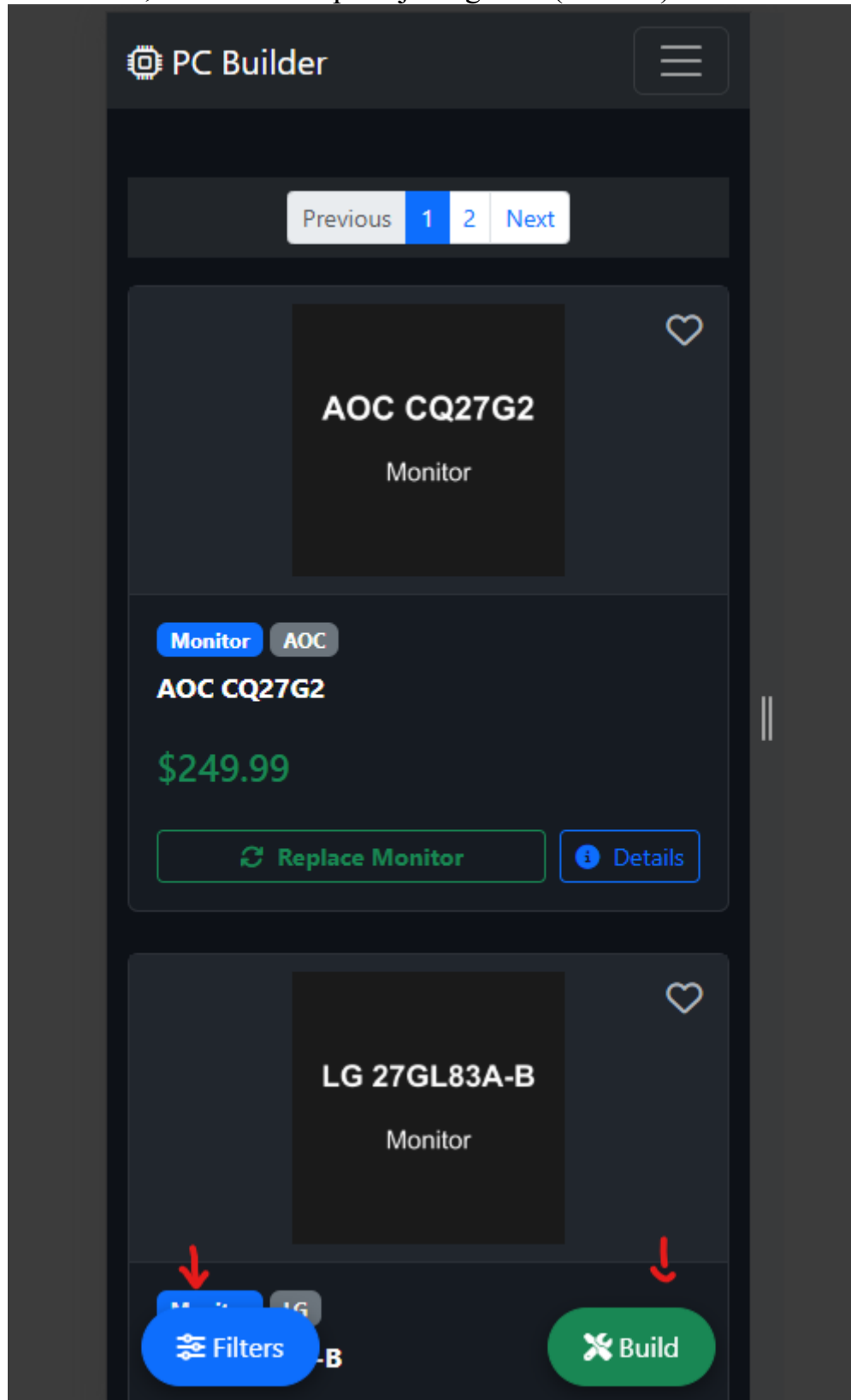
Slika 6. Saved builds

Kada upišemo u *Build name* ime (Slika 6. ja sam upisao primer) i kliknemo *Save Build*,

build postaje dostupan u *Saved Builds*. Klikom na *Load*, možemo da uvezemo build i zamenimo ga sa trenutnim build-om. Klikom na ikonicu kante, brišemo build iz local storage-a.

2.2. Pregled sajta [Vizuelni pregled (uredjaji < 768px)]

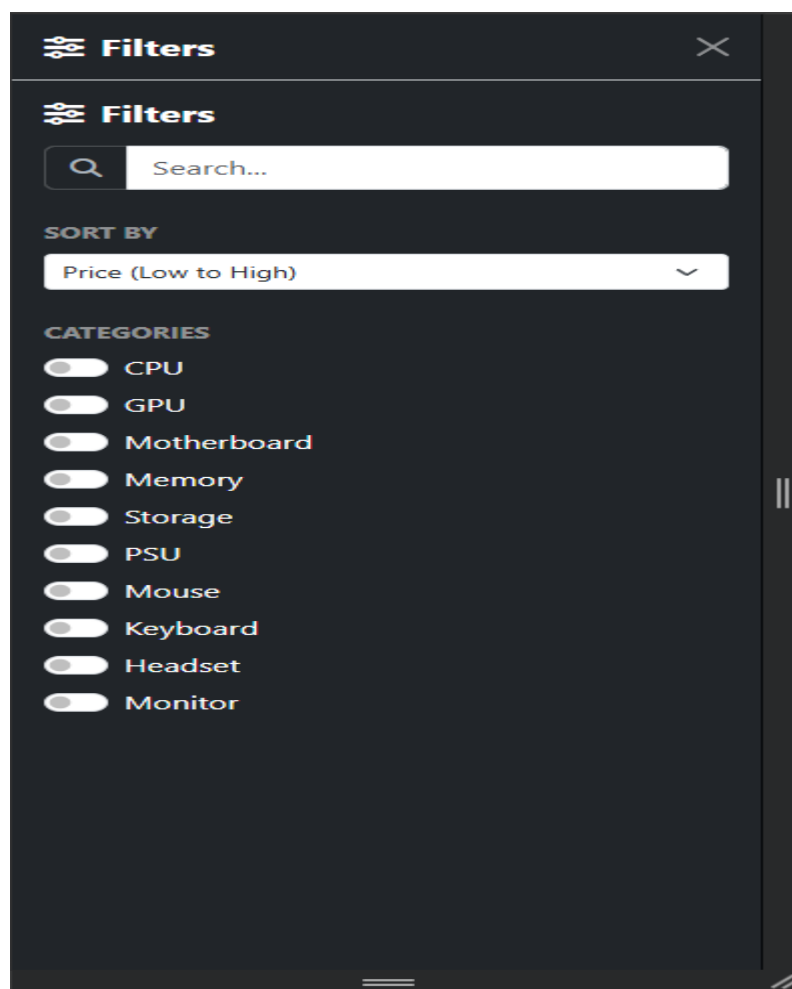
Kada smo na telefonu, filters i build postaju dugmad. (Slika 7.)



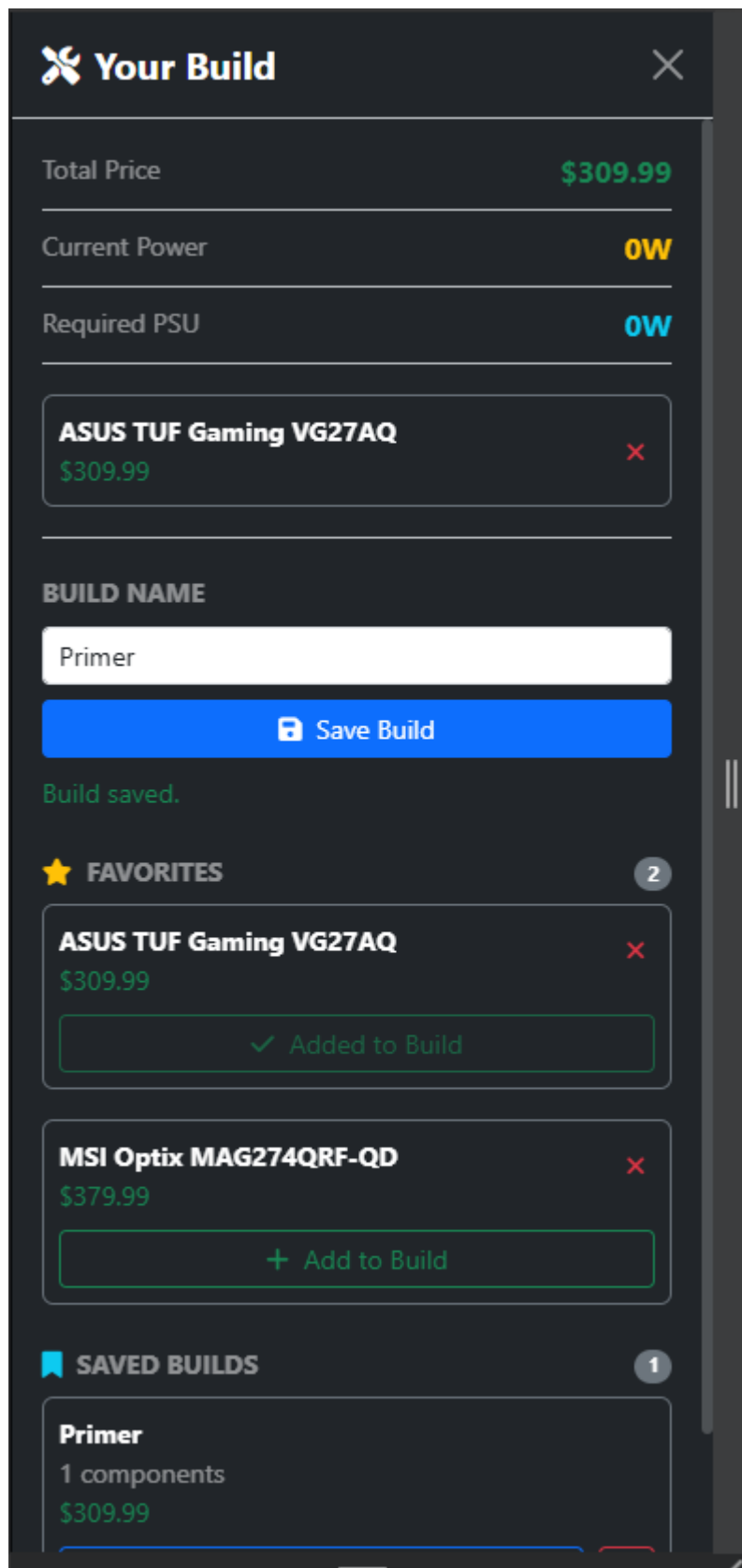
Slika 7. Filters i Build na mobilnom

Klikom na filters, otvara se bootstrap offcanvas za filtere (Slika 8.)

Klikom na build, otvara se bootstrap offcanvas za build (Slika 9.)



Slika 8. bootstrap offcanvas za filtere



Slika 9. bootstrap offcanvas za build

suggest.html:

PC Builder

Home

Builder

Suggest Component

About Author

Documentation

Suggest a Component

Have a component you'd like to see in our database? Fill out the form below with the details. Our team will review your suggestion.

Component Category *

Select Category

Component Name *

e.g., AMD Ryzen 9 7950X

Brand

e.g., AMD, Intel, NVIDIA, ASUS

Price (USD)

e.g., 599.99

Release Date

mm/dd/yyyy

Key Specifications

Enter key specs like cores, clock speed, memory, etc.

Your Email *

your.email@example.com

Clear

Submit Suggestion

author.html:



About the Author

 Name
Marko Pešić

 Index
86/23

 Email
marko.pesic.86.23@ict.edu.rs

 University
Faculty of Information Technology and Engineering

Technologies Used

-  HTML5
-  CSS3
-  JavaScript
-  Bootstrap 5
-  AJAX
-  JSON

[View Documentation](#) [Back to Home](#)

3.1. Ključne funkcije (opis)

loadComponents: Koristio sam fetch api za dohvatanje komponenti

```
function loadComponents() {  
  fetch(COMPONENTS_URL)  
    .then(response => {  
      if (!response.ok) {  
        throw new Error(`Failed to load components: ${response.status}`);  
      }  
      return response.json();  
    })  
    .then(data => {  
      allComponents = data;  
    })  
}
```

```

        filteredComponents = data;
        loadSavedState();
        applyCategoryFromUrl();
        performFiltering();
        updateBuildDisplay();
    })
    .catch(error => {
        console.error(error);
        const componentsHolder = document.getElementById('components-holder');
        if (componentsHolder) {
            componentsHolder.innerHTML = '<div class="col-12"><div class="alert alert-danger">Components could
not be loaded.</div></div>';
        }
    });
}

```

Dovavanje listenera na DOM. Nakon load-a, pozovi ostale listener-e

Ako postoji div `'components-holder'` onda smo na builder stranici. Ovo sam radio da bi izbegao console greske sobzirom da na drugim stranicama nema komponenti.

```

document.addEventListener('DOMContentLoaded', function () {
    helpers.fixAria();
    helpers.footerText();
    setupEventListeners();

    if (document.getElementById('components-holder')) {
        loadComponents();
    }
});

```

applyCategoryFromUrl:

```

function applyCategoryFromUrl() {
    const category = new URLSearchParams(window.location.search).get('category');
    if (!category) return;

    document.querySelectorAll('.form-check-input[name="category"][value="$
{CSS.escape(category)}"]').forEach(checkbox => {
        checkbox.checked = true;
    });
}

```

ova funkcija služi za onaj deo na početnoj strani kada korisnik može da klikne kategoriju. Automatski mu se selektuje izabrana kategorija u builder-u.

Render components:

```

function renderComponents(components) {

```



```

const componentsHolder = document.getElementById('components-holder');
if (!componentsHolder) return;

if (components.length === 0) {
  componentsHolder.innerHTML = '<div class="col-12"><div class="alert alert-secondary">No components
match your filters.</div></div>';
  return;
}

componentsHolder.innerHTML = components.map(component => {
  const isInBuild = buildComponents.some(comp => comp.id === component.id);
  const sameCategoryComponent = buildComponents.find(comp => comp.category === component.category);
  const willReplace = sameCategoryComponent && sameCategoryComponent.id !== component.id;
  const isFavorite = favoriteComponentIds.includes(component.id);
  const buttonClass = isInBuild
    ? 'btn btn-danger btn-sm flex-grow-1 fw-bold add-to-build-btn'
    : 'btn btn-outline-success btn-sm flex-grow-1 fw-bold add-to-build-btn';
  const buttonText = getBuildButtonText(component, isInBuild, willReplace);
  const favoriteClass = isFavorite ? 'fa-solid fa-heart favorite-icon active' : 'fa-regular fa-heart favorite-icon';

  return `
    <div class="col-12 col-lg-6 col-xl-4">
      <div class="card component-card h-100 position-relative">
        <i class="${favoriteClass}" title="Toggle favorite" data-component-id="${
helpers.escapeHTML(component.id)}"></i>
        <div class="card-img-wrapper">
          
        </div>
        <div class="card-body d-flex flex-column">
          <div class="mb-2 d-flex flex-wrap gap-1">
            <span class="badge bg-primary">${helpers.prettyText(component.category)}</span>
            <span class="badge bg-secondary">${helpers.prettyText(component.brand)}</span>
          </div>
          <h6 class="card-title fw-bold">${helpers.escapeHTML(component.name)}</h6>
          <h4 class="text-success mt-auto pt-3">${helpers.formatPrice(component.price)}</h4>
          <div class="d-flex gap-2 mt-3">
            <button class="${buttonClass}" data-component-id="${helpers.escapeHTML(component.id)}">
              ${buttonText}
            </button>
            <button class="btn btn-outline-primary btn-sm" data-bs-toggle="modal" data-bs-
target="#componentModal" data-component-id="${helpers.escapeHTML(component.id)}">
              <i class="fa-solid fa-circle-info me-1"></i> Details
            </button>
          </div>
        </div>
      </div>
    `;
});

```

```

        </div>
    </div>
</div>
`
);
}).join("");
}

```

renderPaginationControls: Crtanje paginacije

```

function renderPaginationControls(totalPages) {
    const paginationBars = document.querySelectorAll('.pagination');
    if (paginationBars.length === 0) return;

    if (totalPages <= 1) {
        paginationBars.forEach(pagination => {
            pagination.innerHTML = "";
        });
        return;
    }

    let paginationHTML = "";

    paginationHTML += currentPage === 1
        ? '<li class="page-item disabled"><span class="page-link">Previous</span></li>'
        : '<li class="page-item"><a class="page-link" href="#" data-page="{currentPage - 1}">Previous</a></li>';

    getPagesToShow(currentPage, totalPages).forEach(page => {
        if (page === '...') {
            paginationHTML += '<li class="page-item disabled"><span class="page-link">...</span></li>';
        } else if (page === currentPage) {
            paginationHTML += '<li class="page-item active"><span class="page-link">{page}</span></li>';
        } else {
            paginationHTML += '<li class="page-item"><a class="page-link" href="#" data-page="{page}">{page}</a></li>';
        }
    });

    paginationHTML += currentPage === totalPages
        ? '<li class="page-item disabled"><span class="page-link">Next</span></li>'
        : '<li class="page-item"><a class="page-link" href="#" data-page="{currentPage + 1}">Next</a></li>';

    paginationBars.forEach(pagination => {
        pagination.innerHTML = paginationHTML;
    });

    document.querySelectorAll('.pagination .page-link[data-page]').forEach(link => {

```

```

link.addEventListener('click', function (event) {
  event.preventDefault();
  currentPage = Number(this.dataset.page);
  renderWithPagination();
  window.scrollTo({ top: 0, behavior: 'smooth' });
});
});
}

```

populateModal: modal je jedan a popunjavam ga sa info-om svake komponente.

```

function populateModal(component) {
  const modalTitle = document.querySelector('#componentModal .modal-title');
  const modalDisplayTitle = document.querySelector('#modal-title');
  const modalImage = document.querySelector('#modal-image');
  const modalSpecs = document.querySelector('#modal-specs');
  const modalPrice = document.querySelector('#modal-price');
  const modalCategory = document.querySelector('#modal-category');
  const modalBrand = document.querySelector('#modal-brand');
  const modalBuildButton = document.querySelector('#modal-build-button');

  if (modalTitle) modalTitle.textContent = component.name;
  if (modalDisplayTitle) modalDisplayTitle.textContent = component.name;
  if (modalImage) {
    modalImage.src = helpers.normalizeImagePath(component.image_path);
    modalImage.alt = component.name;
  }
  if (modalPrice) modalPrice.textContent = helpers.formatPrice(component.price);
  if (modalCategory) modalCategory.textContent = helpers.prettyText(component.category);
  if (modalBrand) modalBrand.textContent = helpers.prettyText(component.brand);
  if (modalSpecs) modalSpecs.innerHTML = formatComponentSpecsRecursive(component.specs || {});

  if (modalBuildButton) {
    const isInBuild = buildComponents.some(comp => comp.id === component.id);
    const sameCategoryComponent = buildComponents.find(comp => comp.category === component.category);
    const willReplace = sameCategoryComponent && sameCategoryComponent.id !== component.id;
    modalBuildButton.dataset.componentId = component.id;
    modalBuildButton.className = getModalBuildButtonClass(isInBuild, willReplace);
    modalBuildButton.innerHTML = getModalBuildButtonText(component, isInBuild, willReplace);
  }
}

```

formatComponentSpecsRecursive: Ovo je gorepomenuta rekurzivna f-j-a koja poziva sebe ako je objekat ugnježen i sadrži još objekata.

```

function formatComponentSpecsRecursive(specs, level = 0) {

```

```

return Object.entries(specs).map(([key, value]) => {
  const prettyKey = helpers.prettyText(key);

  if (typeof value === 'object' && value !== null && !Array.isArray(value)) {
    return `
      <div class="mb-3 ${level > 0 ? 'ps-3' : ''}">
        <h6 class="text-uppercase fw-bold ${level === 0 ? 'text-primary' : 'text-secondary'}">${
        prettyKey}</h6>
        <div class="${level > 0 ? 'ps-3' : ''}">
          ${formatComponentSpecsRecursive(value, level + 1)}
        </div>
      </div>
    `;
  }

  const displayValue = Array.isArray(value) ? value.join(', ') : value;
  return `
    <div class="d-flex justify-content-between gap-3 py-2 border-bottom ${level > 0 ? 'ps-3' : ''}">
      <span class="text-muted">${prettyKey}</span>
      <span class="fw-medium text-end">${helpers.escapeHTML(String(displayValue))}</span>
    </div>
  `;
}).join("");
}

```

addOrReplaceBuildComponent: Ako postoji, menjamo je, ne dodajemo opet.

```

function addOrReplaceBuildComponent(component) {
  buildComponents = buildComponents.filter(existingComponent => existingComponent.category !==
  component.category);
  buildComponents.push(component);
}

```

renderCompatibilityAlerts: dodajemo poruku korisniku ako mu build nije dobar (više od jedne komponente, socket se ne poklapa (matična i procesor), ram se ne poklapa sa matičnom, napajanje nije dovoljno jako za izabrani build, skladište se ne poklapa sa matičnom)

```

function renderCompatibilityAlerts(containerId) {
  const container = document.getElementById(containerId);
  if (!container) return;
}

```

```

const alerts = getCompatibilityAlerts();
if (alerts.length === 0) {
  container.innerHTML = "";
  return;
}

container.innerHTML = alerts.map(alert => `
  <div class="alert alert-${alert.type} py-2 px-2 small mb-2">
    <i class="fa-solid fa-triangle-exclamation me-1"></i>${helpers.escapeHTML(alert.message)}
  </div>
`).join("");
}

```

getCompatibilityAlerts:

```

function getCompatibilityAlerts() {
  const alerts = [];
  const processor = buildComponents.find(component => component.category === 'processor');
  const motherboard = buildComponents.find(component => component.category === 'motherboard');
  const memory = buildComponents.find(component => component.category === 'memory');
  const storage = buildComponents.find(component => component.category === 'storage');
  const powerSupply = buildComponents.find(component => component.category === 'power_supply');
  const power = calculatePowerSummary();

  if (processor && motherboard) {
    const cpuSocket = processor.specs?.socket;
    const motherboardSocket = motherboard.specs?.socket;
    if (cpuSocket && motherboardSocket && cpuSocket !== motherboardSocket) {
      alerts.push({
        type: 'danger',
        message: `CPU socket ${cpuSocket} does not match motherboard socket ${motherboardSocket}.`
      });
    }
  }

  if (motherboard && memory) {
    const motherboardMemoryType = motherboard.specs?.memory_type;
    const memoryType = memory.specs?.type;
    if (motherboardMemoryType && memoryType && motherboardMemoryType !== memoryType) {
      alerts.push({
        type: 'danger',
        message: `Motherboard supports ${motherboardMemoryType}, but selected RAM is ${memoryType}.`
      });
    }
  }
}

```

```

    }

    if (motherboard && storage) {
        const storageSupport = getStorageCompatibility(motherboard, storage);
        if (!storageSupport.compatible) {
            alerts.push({
                type: 'danger',
                message: storageSupport.message
            });
        }
    }

    if (powerSupply) {
        const psuWattage = helpers.extractPsuWattage(powerSupply);
        if (psuWattage > 0 && psuWattage < power.requiredPower) {
            alerts.push({
                type: 'warning',
                message: `Selected PSU capacity is ${psuWattage}W. Estimated draw is ${power.currentPower}W;
recommended PSU is ${power.requiredPower}W.`
            });
        }
    }

    return alerts;
}

```

4.1. Ceo kod.

4.1.1. HTML

index.html:

```

<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description"
        content="PC Builder - Build your custom PC with ease. Choose components, compare prices, and create your
dream build.">
    <title>PC Builder - Home</title>
    <!-- Bootstrap 5 CSS -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
    <!-- Bootstrap Icons -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.1/font/bootstrap-icons.css" rel="stylesheet">
    <!-- Font Awesome Icons -->
    <link href="https://cdn.jsdelivr.net/npm/font-awesome@6.5.2/css/all.min.css" rel="stylesheet">
    <!-- Custom CSS -->

```

```

<link rel="stylesheet" href="assets/css/style.css">
<!-- Favicon -->
<link rel="shortcut icon" href="assets/images/favicon/fav.png" type="image/x-icon">
</head>

<body>
<!-- Navigation -->
<nav class="navbar navbar-expand-lg navbar-dark bg-dark fixed-top">
  <div class="container">
    <a class="navbar-brand" href="index.html">
      <i class="bi bi-cpu"></i> PC Builder
    </a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarNav">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item">
          <a class="nav-link active" href="index.html"><i class="bi bi-house"></i> Home</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="builder.html"><i class="bi bi-tools"></i> Builder</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="suggest.html"><i class="bi bi-plus-circle"></i> Suggest Component</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="author.html"><i class="bi bi-person"></i> About Author</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="assets/other/doc.pdf" target="_blank"><i class="bi bi-book"></i>
            Documentation</a>
        </li>
      </ul>
    </div>
  </div>
</nav>

<!-- Hero Section -->
<header class="hero-section bg-dark text-white py-5" style="margin-top: 56px;">
  <div class="container">
    <div class="row align-items-center min-vh-50">
      <div class="col-lg-6">
        <h1 class="display-4 fw-bold">Build Your Dream PC</h1>
        <p class="lead">Choose from the best components, compare prices, and create the perfect custom PC

```

```

        build for your needs.</p>
        <a href="builder.html" class="btn btn-primary btn-lg">
          <i class="bi bi-tools"></i> Start Building
        </a>
      </div>
      <div class="col-lg-6 text-center">
        <i class="bi bi-pc-display" style="font-size: 10rem;"></i>
      </div>
    </div>
  </div>
</header>

<!-- Features Section -->
<section class="py-5">
  <div class="container">
    <h2 class="text-center text-white mb-5">Why Use Our PC Builder?</h2>
    <div class="row g-4">
      <div class="col-md-4">
        <div class="card h-100 border-0 shadow-sm">
          <div class="card-body text-center">
            <i class="bi bi-search display-4 text-primary mb-3"></i>
            <h5 class="card-title">Easy Component Selection</h5>
            <p class="card-text">Browse and filter through a wide range of PC components with detailed specifications.</p>
          </div>
        </div>
      </div>
      <div class="col-md-4">
        <div class="card h-100 border-0 shadow-sm">
          <div class="card-body text-center">
            <i class="bi bi-calculator display-4 text-success mb-3"></i>
            <h5 class="card-title">Price Calculation</h5>
            <p class="card-text">Real-time price calculation and budget tracking as you build your PC.
          </p>
          </div>
        </div>
      </div>
      <div class="col-md-4">
        <div class="card h-100 border-0 shadow-sm">
          <div class="card-body text-center">
            <i class="bi bi-save display-4 text-info mb-3"></i>
            <h5 class="card-title">Save Your Build</h5>
            <p class="card-text">Save your PC configuration to local storage and load it anytime.</p>
          </div>
        </div>
      </div>
    </div>
  </div>

```



```
        </div>
    </div>
</div>
</section>
```

```
<!-- Categories Preview -->
```

```
<section class="py-5 bg-light">
```

```
    <div class="container">
```

```
        <h2 class="text-center mb-5">Component Categories</h2>
```

```
        <div class="row g-4">
```

```
            <div class="col-6 col-md-4 col-lg-2">
```

```
                <a href="builder.html?category=processor" class="text-decoration-none">
```

```
                    <div class="card category-card h-100 border-0 shadow-sm text-center">
```

```
                        <div class="card-body">
```

```
                            <i class="bi bi-cpu display-5 mb-2"></i>
```

```
                            <h6 class="card-title text-white">Processors</h6>
```

```
                        </div>
```

```
                    </div>
```

```
                </a>
```

```
            </div>
```

```
            <div class="col-6 col-md-4 col-lg-2">
```

```
                <a href="builder.html?category=graphics_card" class="text-decoration-none">
```

```
                    <div class="card category-card h-100 border-0 shadow-sm text-center">
```

```
                        <div class="card-body">
```

```
                            <i class="bi bi-gpu-card display-5 mb-2"></i>
```

```
                            <h6 class="card-title text-white">Graphics Cards</h6>
```

```
                        </div>
```

```
                    </div>
```

```
                </a>
```

```
            </div>
```

```
            <div class="col-6 col-md-4 col-lg-2">
```

```
                <a href="builder.html?category=motherboard" class="text-decoration-none">
```

```
                    <div class="card category-card h-100 border-0 shadow-sm text-center">
```

```
                        <div class="card-body">
```

```
                            <i class="bi bi-motherboard display-5 mb-2"></i>
```

```
                            <h6 class="card-title text-white">Motherboards</h6>
```

```
                        </div>
```

```
                    </div>
```

```
                </a>
```

```
            </div>
```

```
            <div class="col-6 col-md-4 col-lg-2">
```

```
                <a href="builder.html?category=memory" class="text-decoration-none">
```

```
                    <div class="card category-card h-100 border-0 shadow-sm text-center">
```

```
                        <div class="card-body">
```

```
                            <i class="bi bi-memory display-5 mb-2"></i>
```

```
<h6 class="card-title text-white">Memory</h6>  
</div>  
</div>  
</a>  
</div>  
<div class="col-6 col-md-4 col-lg-2">  
  <a href="builder.html?category=storage" class="text-decoration-none">  
    <div class="card category-card h-100 border-0 shadow-sm text-center">  
      <div class="card-body">  
        <i class="bi bi-device-hdd display-5 mb-2"></i>  
        <h6 class="card-title text-white">Storage</h6>  
      </div>  
    </div>  
  </a>  
</div>  
<div class="col-6 col-md-4 col-lg-2">  
  <a href="builder.html?category=power_supply" class="text-decoration-none">  
    <div class="card category-card h-100 border-0 shadow-sm text-center">  
      <div class="card-body">  
        <i class="bi bi-plug display-5 mb-2"></i>  
        <h6 class="card-title text-white">Power Supplies</h6>  
      </div>  
    </div>  
  </a>  
</div>  
</div>  
</div>  
</section>  
  
<!-- Footer -->  
<footer class="bg-dark text-white py-4 mt-5">  
  <div class="container">  
    <div class="row">  
      <div class="col-md-6">  
        <h5><i class="bi bi-cpu"></i> PC Builder</h5>  
        <p class="small">Build your custom PC with ease.</p>  
      </div>  
      <div class="col-md-6 text-md-end">  
        <ul class="list-inline mb-0">  
          <li class="list-inline-item"><a href="index.html" class="text-white text-decoration-none">Home</a></li>  
          <li class="list-inline-item"><a href="builder.html" class="text-white text-decoration-none">Builder</a></li>  
          <li class="list-inline-item"><a href="suggest.html" class="text-white text-decoration-none">Suggest  
Component</a></li>
```

```

        <li class="list-inline-item"><a href="author.html" class="text-white text-decoration-none">About
Author</a></li>
        <li class="list-inline-item"><a href="assets/other/doc.pdf" target="_blank" class="text-white text-
decoration-none">Documentation</a></li>
    </ul>
    <p class="small" id="footer-text">&copy; 2026 PC Builder. All rights reserved.</p>
</div>
</div>
</div>
</footer>

<!-- Bootstrap 5 JS Bundle -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
<!-- Custom JS -->
<script src="assets/js/main.js"></script>
</body>

</html>

```

builder.html:

```

<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description"
        content="PC Builder - Build your custom PC with ease. Choose components, compare prices, and create your
dream build.">
    <title>PC Builder - Builder</title>
    <!-- Bootstrap 5 CSS -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
    <!-- Bootstrap Icons -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.1/font/bootstrap-icons.css" rel="stylesheet">
    <!-- Font Awesome Icons -->
    <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.2/css/all.min.css" rel="stylesheet">
    <!-- Custom CSS -->
    <link rel="stylesheet" href="assets/css/style.css">
    <!-- Favicon -->
    <link rel="shortcut icon" href="assets/images/favicon/fav.png" type="image/x-icon">
</head>

<body>

    <!-- ===== NAVIGATION ===== -->

```

```

<nav class="navbar navbar-expand-lg navbar-dark bg-dark fixed-top">
  <div class="container">
    <a class="navbar-brand" href="index.html">
      <i class="bi bi-cpu"></i> PC Builder
    </a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarNav">
      <ul class="navbar-nav ms-auto">
        <li class="nav-item">
          <a class="nav-link" href="index.html"><i class="bi bi-house"></i> Home</a>
        </li>
        <li class="nav-item">
          <a class="nav-link active" href="builder.html"><i class="bi bi-tools"></i> Builder</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="suggest.html"><i class="bi bi-plus-circle"></i> Suggest Component</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="author.html"><i class="bi bi-person"></i> About Author</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="assets/other/doc.pdf" target="_blank"><i class="bi bi-book"></i>
            Documentation</a>
        </li>
      </ul>
    </div>
  </div>
</nav>

```

```

<!-- ===== MOBILE TOGGLE BUTTONS ===== -->

```

```

<!-- Mobile Filter Toggle Button -->

```

```

<button class="btn btn-primary mobile-filter-toggle d-md-none" type="button" data-bs-toggle="offcanvas"
  data-bs-target="#filtersCanvas">
  <i class="fa-solid fa-sliders"></i> <span class="text-white">Filters</span>
</button>

```

```

<!-- Mobile Build Toggle Button -->

```

```

<button class="btn btn-success mobile-build-toggle d-md-none" type="button" data-bs-toggle="offcanvas"
  data-bs-target="#builderCanvas">
  <i class="fa-solid fa-screwdriver-wrench"></i> <span class="text-white">Build</span>
</button>

```

```

<!-- ===== MAIN CONTENT AREA ===== -->

```

```

<div class="container-fluid mt-3 mb-5">
  <div class="row px-md-3">

    <!-- ===== LEFT SIDEBAR - DESKTOP FILTERS ===== -->
    <aside class="col-md-3 col-lg-2 d-none d-md-block sidebar">
      <div class="side-panel">
        <h5 class="fw-bold mb-3 text-white"><i class="fa-solid fa-sliders text-white me-2"></i>Filters</h5>

        <div class="input-group mb-4">
          <span class="input-group-text bg-transparent">
            <i class="fa-solid fa-magnifying-glass"></i>
          </span>
          <input type="text" id="search-desktop" name="search-desktop" class="form-control"
            placeholder="Search..." aria-label="Search components">
        </div>

        <div class="mb-4">
          <label for="sort-desktop" class="form-label text-white-50 small text-uppercase fw-bold">Sort
            By</label>
          <select id="sort-desktop" name="sort-desktop" class="form-select form-select-sm">
            <option value="price-low-high">Price (Low to High)</option>
            <option value="price-high-low">Price (High to Low)</option>
            <option value="a-z">A-Z</option>
            <option value="z-a">Z-A</option>
            <option value="newest">Date Released (Newest First)</option>
            <option value="oldest">Date Released (Oldest First)</option>
          </select>
        </div>

        <div class="mb-3">
          <label class="form-label text-white-50 small text-uppercase fw-bold">Categories</label>

          <div class="form-check form-switch mb-2">
            <input class="form-check-input" type="checkbox" id="cpu-desktop" name="category"
              value="processor">
            <label class="form-check-label" for="cpu-desktop">CPU</label>
          </div>

          <div class="form-check form-switch mb-2">
            <input class="form-check-input" type="checkbox" id="gpu-desktop" name="category"
              value="graphics_card">
            <label class="form-check-label" for="gpu-desktop">GPU</label>
          </div>

          <div class="form-check form-switch mb-2">

```

```
<input class="form-check-input" type="checkbox" id="mobo-desktop" name="category"
      value="motherboard">
<label class="form-check-label" for="mobo-desktop">Motherboard</label>
</div>
```

```
<div class="form-check form-switch mb-2">
  <input class="form-check-input" type="checkbox" id="ram-desktop" name="category"
        value="memory">
  <label class="form-check-label" for="ram-desktop">Memory</label>
</div>
```

```
<div class="form-check form-switch mb-2">
  <input class="form-check-input" type="checkbox" id="storage-desktop" name="category"
        value="storage">
  <label class="form-check-label" for="storage-desktop">Storage</label>
</div>
```

```
<div class="form-check form-switch mb-2">
  <input class="form-check-input" type="checkbox" id="psu-desktop" name="category"
        value="power_supply">
  <label class="form-check-label" for="psu-desktop">PSU</label>
</div>
```

```
<div class="form-check form-switch mb-2">
  <input class="form-check-input" type="checkbox" id="mouse-desktop" name="category"
        value="mouse">
  <label class="form-check-label" for="mouse-desktop">Mouse</label>
</div>
```

```
<div class="form-check form-switch mb-2">
  <input class="form-check-input" type="checkbox" id="keyboard-desktop" name="category"
        value="keyboard">
  <label class="form-check-label" for="keyboard-desktop">Keyboard</label>
</div>
```

```
<div class="form-check form-switch mb-2">
  <input class="form-check-input" type="checkbox" id="headset-desktop" name="category"
        value="headset">
  <label class="form-check-label" for="headset-desktop">Headset</label>
</div>
```

```
<div class="form-check form-switch mb-2">
  <input class="form-check-input" type="checkbox" id="monitor-desktop" name="category"
        value="monitor">
  <label class="form-check-label" for="monitor-desktop">Monitor</label>
```

```

        </div>
    </div>
</div>
</aside>

<!-- ===== MAIN CONTENT - COMPONENTS ===== -->
<main class="col-12 col-md-6 col-lg-8">
    <!-- Top Pagination -->
    <div class="bg-dark pt-2 pb-2 mb-3">
        <div class="container">
            <div class="row">
                <div class="col-12">
                    <div class="d-flex justify-content-center">
                        <nav>
                            <ul class="pagination pagination-sm mb-0">
                                <li class="page-item disabled"><a class="page-link" href="#">Previous</a></li>
                                <li class="page-item active"><a class="page-link" href="#">1</a></li>
                                <li class="page-item"><a class="page-link" href="#">2</a></li>
                                <li class="page-item"><a class="page-link" href="#">3</a></li>
                                <li class="page-item"><a class="page-link" href="#">Next</a></li>
                            </ul>
                        </nav>
                    </div>
                </div>
            </div>
        </div>
    </div>

    <!-- ===== COMPONENTS GRID ===== -->
    <div class="row g-4" id="components-holder">
        <div class="col-12">
            <div class="alert alert-secondary mb-0">Loading components...</div>
        </div>
    </div>

    <!-- ===== BOTTOM PAGINATION ===== -->
    <!-- Bottom Pagination -->
    <div class="bg-dark pt-2 pb-2 mt-3">
        <div class="container">
            <div class="row">
                <div class="col-12">
                    <div class="d-flex justify-content-center">
                        <nav>
                            <ul class="pagination pagination-sm mb-0">
                                <li class="page-item disabled"><a class="page-link" href="#">Previous</a></li>

```

```

        <li class="page-item active"><a class="page-link" href="#">1</a></li>
        <li class="page-item"><a class="page-link" href="#">2</a></li>
        <li class="page-item"><a class="page-link" href="#">3</a></li>
        <li class="page-item"><a class="page-link" href="#">Next</a></li>
    </ul>
</nav>
</div>
</div>
</div>
</div>
</div>
</div>

<!-- ===== RIGHT SIDEBAR - BUILD & FAVORITES ===== -->
<aside class="col-md-3 col-lg-2 d-none d-md-block sidebar">
    <div class="side-panel mb-4" id="desktop-build-panel">
        <h5 class="fw-bold mb-3 text-white"><i
            class="fa-solid fa-screwdriver-wrench text-white me-2"></i>Your Build
        </h5>

        <div class="d-flex justify-content-between border-bottom border-secondary-subtle pb-2 mb-2">
            <span class="small text-white-50">Total Price</span>
            <span class="fw-bold text-success" id="desktop-total-price">$0.00</span>
        </div>
        <div class="d-flex justify-content-between border-bottom border-secondary-subtle pb-2 mb-2">
            <span class="small text-white-50">Current Power</span>
            <span class="fw-bold text-warning" id="desktop-current-power">0W</span>
        </div>
        <div class="d-flex justify-content-between border-bottom border-secondary-subtle pb-2 mb-3">
            <span class="small text-white-50">Required PSU</span>
            <span class="fw-bold text-info" id="desktop-required-power">0W</span>
        </div>

        <div id="desktop-compatibility-alerts"></div>

        <div id="desktop-build-list"></div>

        <div class="mt-3 pt-3 border-top border-secondary-subtle">
            <label for="desktop-build-name" class="form-label text-white-50 small text-uppercase fw-
                bold">Build Name</label>
            <input type="text" class="form-control form-control-sm build-name-input" id="desktop-build-name"
                placeholder="Gaming build">
            <button class="btn btn-primary btn-sm w-100 mt-2 save-build-btn" type="button">

```



```

        <i class="fa-solid fa-floppy-disk me-1"></i> Save Build
    </button>
    <p class="small mt-2 mb-0 save-build-status text-white-50"></p>
</div>

</div>

<div class="side-panel" id="desktop-favorites-panel">
    <h6 class="fw-bold text-uppercase small text-white-50 mb-2"><i
        class="fa-solid fa-star text-warning me-2"></i> Favorites
    <span class="badge bg-secondary rounded-pill float-end" id="desktop-favorite-count">0</span>
</h6>
    <div id="desktop-favorites-list"></div>

    <h6 class="fw-bold text-uppercase small text-white-50 mt-4 mb-2"><i
        class="fa-solid fa-bookmark text-info me-2"></i> Saved Builds
    <span class="badge bg-secondary rounded-pill float-end" id="desktop-saved-build-count">0</span>
</h6>
    <div id="desktop-saved-builds-list"></div>
</div>
</aside>

</div>
</div>

<!-- ===== MOBILE FILTERS OFFCANVAS ===== -->
<div class="offcanvas offcanvas-start bg-dark" id="filtersCanvas">
    <div class="offcanvas-header border-bottom border-secondary-subtle">
        <h5 class="fw-bold mb-0 text-white"><i class="fa-solid fa-sliders text-white me-2"></i> Filters</h5>
        <button class="btn-close btn-close-white" data-bs-dismiss="offcanvas" aria-label="Close"></button>
    </div>
    <div class="offcanvas-body">
        <h5 class="fw-bold mb-3 text-white"><i class="fa-solid fa-sliders text-white me-2"></i> Filters</h5>

        <div class="input-group mb-4">
            <span class="input-group-text bg-transparent"><i class="fa-solid fa-magnifying-glass"></i></span>
            <input type="text" id="search-mobile" name="search-mobile" class="form-control"
placeholder="Search..."
            aria-label="Search components">
        </div>

        <div class="mb-4">
            <label for="sort-mobile" class="form-label text-white-50 small text-uppercase fw-bold">Sort By</label>
            <select id="sort-mobile" name="sort-mobile" class="form-select form-select-sm">
                <option value="price-low-high">Price (Low to High)</option>
            </select>
        </div>
    </div>
</div>

```

```

        <option value="price-high-low">Price (High to Low)</option>
        <option value="a-z">A-Z</option>
        <option value="z-a">Z-A</option>
        <option value="newest">Date Released (Newest First)</option>
        <option value="oldest">Date Released (Oldest First)</option>
    </select>
</div>

<div class="mb-3">
    <label class="form-label text-white-50 small text-uppercase fw-bold">Categories</label>

    <div class="form-check form-switch mb-2">
        <input class="form-check-input" type="checkbox" id="cpu-mobile" name="category"
value="processor">
        <label class="form-check-label" for="cpu-mobile">CPU</label>
    </div>

    <div class="form-check form-switch mb-2">
        <input class="form-check-input" type="checkbox" id="gpu-mobile" name="category"
        value="graphics_card">
        <label class="form-check-label" for="gpu-mobile">GPU</label>
    </div>

    <div class="form-check form-switch mb-2">
        <input class="form-check-input" type="checkbox" id="mobo-mobile" name="category"
        value="motherboard">
        <label class="form-check-label" for="mobo-mobile">Motherboard</label>
    </div>

    <div class="form-check form-switch mb-2">
        <input class="form-check-input" type="checkbox" id="ram-mobile" name="category"
value="memory">
        <label class="form-check-label" for="ram-mobile">Memory</label>
    </div>

    <div class="form-check form-switch mb-2">
        <input class="form-check-input" type="checkbox" id="storage-mobile" name="category"
value="storage">
        <label class="form-check-label" for="storage-mobile">Storage</label>
    </div>

    <div class="form-check form-switch mb-2">
        <input class="form-check-input" type="checkbox" id="psu-mobile" name="category"
        value="power_supply">
        <label class="form-check-label" for="psu-mobile">PSU</label>
    </div>

```

```

    </div>

    <div class="form-check form-switch mb-2">
      <input class="form-check-input" type="checkbox" id="mouse-mobile" name="category"
value="mouse">
      <label class="form-check-label" for="mouse-mobile">Mouse</label>
    </div>

    <div class="form-check form-switch mb-2">
      <input class="form-check-input" type="checkbox" id="keyboard-mobile" name="category"
      value="keyboard">
      <label class="form-check-label" for="keyboard-mobile">Keyboard</label>
    </div>

    <div class="form-check form-switch mb-2">
      <input class="form-check-input" type="checkbox" id="headset-mobile" name="category"
value="headset">
      <label class="form-check-label" for="headset-mobile">Headset</label>
    </div>

    <div class="form-check form-switch mb-2">
      <input class="form-check-input" type="checkbox" id="monitor-mobile" name="category"
value="monitor">
      <label class="form-check-label" for="monitor-mobile">Monitor</label>
    </div>
  </div>
</div>

<!-- ===== MOBILE BUILD OFFCANVAS ===== -->
<div class="offcanvas offcanvas-end bg-dark" id="builderCanvas">
  <div class="offcanvas-header border-bottom border-secondary-subtle">
    <h5 class="fw-bold mb-0 text-white"><i class="fa-solid fa-screwdriver-wrench text-white me-2"></i>Your
Build
    </h5>
    <button class="btn-close btn-close-white" data-bs-dismiss="offcanvas" aria-label="Close"></button>
  </div>
  <div class="offcanvas-body">
    <div class="d-flex justify-content-between border-bottom border-secondary-subtle pb-2 mb-2">
      <span class="small text-white-50">Total Price</span>
      <span class="fw-bold text-success" id="mobile-total-price">$0.00</span>
    </div>
    <div class="d-flex justify-content-between border-bottom border-secondary-subtle pb-2 mb-2">
      <span class="small text-white-50">Current Power</span>
      <span class="fw-bold text-warning" id="mobile-current-power">0W</span>

```

```

</div>
<div class="d-flex justify-content-between border-bottom border-secondary-subtle pb-2 mb-3">
  <span class="small text-white-50">Required PSU</span>
  <span class="fw-bold text-info" id="mobile-required-power">0W</span>
</div>

<div id="mobile-compatibility-alerts"></div>

<div id="mobile-build-list"></div>

<div class="mt-3 pt-3 border-top border-secondary-subtle">
  <label for="mobile-build-name" class="form-label text-white-50 small text-uppercase fw-bold">Build
Name</label>
  <input type="text" class="form-control form-control-sm build-name-input" id="mobile-build-name"
placeholder="Gaming build">
  <button class="btn btn-primary btn-sm w-100 mt-2 save-build-btn" type="button">
    <i class="fa-solid fa-floppy-disk me-1"></i> Save Build
  </button>
  <p class="small mt-2 mb-0 save-build-status text-white-50"></p>
</div>

<h6 class="fw-bold text-uppercase small text-white-50 mt-4 mb-2"><i
class="fa-solid fa-star text-warning me-2"></i>Favorites
  <span class="badge bg-secondary rounded-pill float-end" id="mobile-favorite-count">0</span>
</h6>
<div id="mobile-favorites-list"></div>

<h6 class="fw-bold text-uppercase small text-white-50 mt-4 mb-2"><i
class="fa-solid fa-bookmark text-info me-2"></i>Saved Builds
  <span class="badge bg-secondary rounded-pill float-end" id="mobile-saved-build-count">0</span>
</h6>
<div id="mobile-saved-builds-list"></div>
</div>
</div>

<!-- ===== COMPONENT DETAILS MODAL ===== -->
<div class="modal fade" id="componentModal">
  <div class="modal-dialog modal-lg modal-dialog-centered">
    <div class="modal-content">
      <div class="modal-header border-bottom border-secondary-subtle">
        <h5 class="modal-title fw-bold">Component Details</h5>
        <button class="btn-close" data-bs-dismiss="modal" aria-label="Close"></button>
      </div>
      <div class="modal-body">

```

```

<div class="row">
  <div class="col-md-5">
    
    </div>
    <div class="col-md-7">
      <h4 id="modal-title">AMD Ryzen 5 7600X</h4>
      <h3 id="modal-price" class="text-success mb-3">$229.00</h3>

      <!-- Category and Brand Badges -->
      <div class="mb-3 d-flex gap-2">
        <span id="modal-category" class="badge bg-primary">Category</span>
        <span id="modal-brand" class="badge bg-secondary">Brand</span>
      </div>

      <button class="btn btn-success fw-bold" id="modal-build-button"><i class="fa-solid fa-plus
me-1"></i> Add to
        Build</button>
      </div>
    </div>

    <!-- Specifications Section -->
    <div class="row mt-4">
      <div class="col-12">
        <h5 class="fw-bold mb-3 border-bottom pb-2">Specifications</h5>
        <div id="modal-specs">
          <!-- Specs will be dynamically inserted here -->
        </div>
      </div>
    </div>
  </div>
</div>

<!-- ===== SCRIPTS ===== -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"></script>
<script src="assets/js/main.js"></script>
</body>

</html>

```

author.html:

```

<!DOCTYPE html>
<html lang="en">

```

```

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
    content="PC Builder - Build your custom PC with ease. Choose components, compare prices, and create your
dream build.">
  <title>PC Builder - Author</title>
  <!-- Bootstrap 5 CSS -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
  <!-- Bootstrap Icons -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.1/font/bootstrap-icons.css" rel="stylesheet">
  <!-- Font Awesome Icons -->
  <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.2/css/all.min.css" rel="stylesheet">
  <!-- Custom CSS -->
  <link rel="stylesheet" href="assets/css/style.css">
  <!-- Favicon -->
  <link rel="shortcut icon" href="assets/images/favicon/fav.png" type="image/x-icon">
</head>

<body>
  <!-- Navigation -->
  <nav class="navbar navbar-expand-lg navbar-dark bg-dark fixed-top">
    <div class="container">
      <a class="navbar-brand" href="index.html">
        <i class="bi bi-cpu"></i> PC Builder
      </a>
      <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav">
        <span class="navbar-toggler-icon"></span>
      </button>
      <div class="collapse navbar-collapse" id="navbarNav">
        <ul class="navbar-nav ms-auto">
          <li class="nav-item">
            <a class="nav-link" href="index.html"><i class="bi bi-house"></i> Home</a>
          </li>
          <li class="nav-item">
            <a class="nav-link" href="builder.html"><i class="bi bi-tools"></i> Builder</a>
          </li>
          <li class="nav-item">
            <a class="nav-link" href="suggest.html"><i class="bi bi-plus-circle"></i> Suggest Component</a>
          </li>
          <li class="nav-item">
            <a class="nav-link active" href="author.html"><i class="bi bi-person"></i> About Author</a>
          </li>
          <li class="nav-item">

```

```

        <a class="nav-link" href="assets/other/doc.pdf" target="_blank"><i class="bi bi-book"></i>
        Documentation</a>
    </li>
</ul>
</div>
</div>
</nav>

<div class="container mt-4">
    <div class="row justify-content-center">
        <div class="col-lg-8">
            <div class="card shadow">
                <div class="card-body p-5">
                    <div class="row align-items-center">
                        <div class="col-md-4 text-center mb-4 mb-md-0">
                            <div class="author-photo-container">
                                
                            </div>
                        </div>
                        <div class="col-md-8">
                            <h2 class="mb-3">About the Author</h2>

                            <div class="list-group list-group-flush bg-transparent">
                                <div class="list-group-item bg-transparent border-secondary-subtle py-2 px-0">
                                    <small class="text-white-50"><i class="bi bi-person-fill me-2"></i>Name</small>
                                    <div class="fw-bold"><span class="text-light">Marko Pešić</span></div>
                                </div>
                                <div class="list-group-item bg-transparent border-secondary-subtle py-2 px-0">
                                    <small class="text-white-50"><i class="bi bi-hash me-2"></i>Index</small>
                                    <div class="fw-bold"><span class="text-light">86/23</span></div>
                                </div>
                                <div class="list-group-item bg-transparent border-secondary-subtle py-2 px-0">
                                    <small class="text-white-50"><i class="bi bi-envelope me-2"></i>Email</small>
                                    <div class="fw-bold text-break"><span class="text-
light">marko.pesic.86.23@ict.edu.rs</span></div>
                                </div>
                                <div class="list-group-item bg-transparent py-2 px-0">
                                    <small class="text-white-50"><i class="bi bi-building me-2"></i>University</small>
                                    <div class="fw-bold"><span class="text-light">Faculty of Information Technology and
Engineering</span></div>
                                </div>
                            </div>
                        </div>
                    </div>
                </div>
            </div>
        </div>
    </div>
</div>

```

<hr>

<h5>Technologies Used</h5>

<p class="text-white-50">

<ul class="list-unstyled">

<li class="list-group-item"><i class="bi bi-code-slash"></i> HTML5

<li class="list-group-item"><i class="bi bi-braces"></i> CSS3

<li class="list-group-item"><i class="bi bi-code"></i> JavaScript

<li class="list-group-item"><i class="bi bi-bootstrap"></i> Bootstrap 5

<li class="list-group-item"><i class="bi bi-arrow-repeat"></i> AJAX

<li class="list-group-item"><i class="bi bi-filetype-json"></i> JSON

</p>

<div class="mt-4 d-flex flex-wrap gap-2">

<i class="bi bi-book"></i> View Documentation

<i class="bi bi-house"></i> Back to Home

</div>

</div>

</div>

</div>

</div>

</div>

</div>

</div>

<!-- Footer -->

<footer class="bg-dark text-white py-4 mt-5">

<div class="container">

<div class="row">

<div class="col-md-6">

<h5><i class="bi bi-cpu"></i> PC Builder</h5>

<p class="small">Build your custom PC with ease.</p>

</div>

<div class="col-md-6 text-md-end">

<ul class="list-inline mb-0">

<li class="list-inline-item"><a href="index.html" class="text-white text-decoration-
none">Home

<li class="list-inline-item"><a href="builder.html" class="text-white text-decoration-
none">Builder


```

        <li class="list-inline-item"><a href="suggest.html" class="text-white text-decoration-none">Suggest
Component</a></li>
        <li class="list-inline-item"><a href="author.html" class="text-white text-decoration-none">About
Author</a></li>
        <li class="list-inline-item"><a href="assets/other/doc.pdf" target="_blank" class="text-white text-
decoration-none">Documentation</a></li>
    </ul>
    <p class="small" id="footer-text">&copy; 2026 PC Builder. All rights reserved.</p>
</div>
</div>
</div>
</footer>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
<script src="assets/js/main.js"></script>
</body>

</html>

```

suggest.html:

```

<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta name="description"
        content="PC Builder - Build your custom PC with ease. Choose components, compare prices, and create your
dream build.">
    <title>PC Builder - Suggest a Component</title>
    <!-- Bootstrap 5 CSS -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
    <!-- Bootstrap Icons -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.1/font/bootstrap-icons.css" rel="stylesheet">
    <!-- Font Awesome Icons -->
    <link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.2/css/all.min.css" rel="stylesheet">
    <!-- Custom CSS -->
    <link rel="stylesheet" href="assets/css/style.css">
    <!-- Favicon -->
    <link rel="shortcut icon" href="assets/images/favicon/fav.png" type="image/x-icon">
</head>
<body>
    <!-- Navigation -->
    <nav class="navbar navbar-expand-lg navbar-dark bg-dark fixed-top">

```

```

<div class="container">
  <a class="navbar-brand" href="index.html">
    <i class="bi bi-cpu"></i> PC Builder
  </a>
  <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav">
    <span class="navbar-toggler-icon"></span>
  </button>
  <div class="collapse navbar-collapse" id="navbarNav">
    <ul class="navbar-nav ms-auto">
      <li class="nav-item">
        <a class="nav-link" href="index.html"><i class="bi bi-house"></i> Home</a>
      </li>
      <li class="nav-item">
        <a class="nav-link" href="builder.html"><i class="bi bi-tools"></i> Builder</a>
      </li>
      <li class="nav-item">
        <a class="nav-link active" href="suggest.html"><i class="bi bi-plus-circle"></i> Suggest
Component</a>
      </li>
      <li class="nav-item">
        <a class="nav-link" href="author.html"><i class="bi bi-person"></i> About Author</a>
      </li>
      <li class="nav-item">
        <a class="nav-link" href="assets/other/doc.pdf" target="_blank"><i class="bi bi-book"></i>
Documentation</a>
      </li>
    </ul>
  </div>
</div>
</nav>

```

```

<div class="container" style="margin-top: 100px;">
  <div class="row justify-content-center">
    <div class="col-lg-8">
      <div class="card shadow">
        <div class="card-header bg-primary text-white">
          <h3 class="mb-0"><i class="bi bi-plus-circle"></i> Suggest a Component</h3>
        </div>
        <div class="card-body">
          <p class="mb-4 text-white">
            Have a component you'd like to see in our database? Fill out the form below with the
            details.
            Our team will review your suggestion.
          </p>

```

```

<form id="suggestForm" novalidate>
  <div id="formMessage" class="alert d-none" role="alert"></div>

  <!-- Component Category -->
  <div class="mb-3">
    <label for="componentCategory" class="form-label">
      <i class="bi bi-tag"></i> Component Category *
    </label>
    <select class="form-select" id="componentCategory" required>
      <option value="">Select Category</option>
      <option value="processor">Processor (CPU)</option>
      <option value="graphics_card">Graphics Card (GPU)</option>
      <option value="motherboard">Motherboard</option>
      <option value="memory">Memory (RAM)</option>
      <option value="storage">Storage</option>
      <option value="power_supply">Power Supply (PSU)</option>
      <option value="cpu_cooler">CPU Cooler</option>
      <option value="case">PC Case</option>
    </select>
    <div class="invalid-feedback">Please select a category.</div>
  </div>

  <!-- Component Name -->
  <div class="mb-3">
    <label for="componentName" class="form-label">
      <i class="bi bi-card-heading"></i> Component Name *
    </label>
    <input type="text" class="form-control" id="componentName"
      placeholder="e.g., AMD Ryzen 9 7950X" required pattern="^[A-Za-z0-9\s\-\.\(\)+/]+$"
      minlength="3" maxlength="80">
    <div class="invalid-feedback">Please enter a valid component name (letters, numbers,
      spaces, hyphens, parentheses only).</div>
  </div>

  <!-- Brand -->
  <div class="mb-3">
    <label for="componentBrand" class="form-label">
      <i class="bi bi-building"></i> Brand
    </label>
    <input type="text" class="form-control" id="componentBrand"
      placeholder="e.g., AMD, Intel, NVIDIA, ASUS" pattern="^[A-Za-z0-9\s\-\.\&]+$"
      maxlength="40">
    <div class="invalid-feedback">Please enter a valid brand name.</div>
  </div>

```

```
<!-- Price -->
<div class="mb-3">
  <label for="componentPrice" class="form-label">
    <i class="bi bi-currency-dollar"></i> Price (USD)
  </label>
  <input type="number" class="form-control" id="componentPrice" placeholder="e.g., 599.99"
    min="0.01" max="9999.99" step="0.01">
  <div class="invalid-feedback">Please enter a valid price.</div>
</div>
```

```
<!-- Release Date -->
<div class="mb-3">
  <label for="releaseDate" class="form-label">
    <i class="bi bi-calendar"></i> Release Date
  </label>
  <input type="date" class="form-control" id="releaseDate">
  <div class="invalid-feedback">Please select a release date.</div>
</div>
```

```
<!-- Specifications -->
<div class="mb-3">
  <label for="componentSpecs" class="form-label">
    <i class="bi bi-list-check"></i> Key Specifications
  </label>
  <textarea class="form-control" id="componentSpecs" rows="4"
    placeholder="Enter key specs like cores, clock speed, memory, etc."
    minlength="10" maxlength="500"></textarea>
  <div class="form-text">Optional, 10-500 characters if provided</div>
  <div class="invalid-feedback">Please provide specifications (min 10 chars).</div>
</div>
```

```
<!-- Email -->
<div class="mb-3">
  <label for="userEmail" class="form-label">
    <i class="bi bi-envelope"></i> Your Email *
  </label>
  <input type="email" class="form-control" id="userEmail"
    placeholder="your.email@example.com" required>
  <div class="form-text">We'll notify you when your suggestion is reviewed.</div>
  <div class="invalid-feedback">Please enter a valid email address.</div>
</div>
```

```
<!-- Submit Buttons -->
<div class="d-grid gap-2 d-md-flex justify-content-md-end">
  <button type="reset" class="btn btn-outline-secondary">
```

```

        <i class="bi bi-x-circle"></i> Clear
    </button>
    <button type="submit" class="btn btn-primary">
        <i class="bi bi-send"></i> Submit Suggestion
    </button>
</div>
</form>

<!-- Success Message (Hidden by default) -->
<div id="successMessage" class="alert alert-success mt-4 d-none">
    <h5><i class="bi bi-check-circle"></i> Thank you!</h5>
    <p>Your component suggestion has been submitted successfully. We'll review it and get back
        to you soon.</p>
    <button class="btn btn-outline-success btn-sm" id="newSuggestion">Submit Another</button>
</div>
</div>
</div>
</div>
</div>
</div>

<!-- Footer -->
<footer class="bg-dark text-white py-4 mt-5">
    <div class="container">
        <div class="row">
            <div class="col-md-6">
                <h5><i class="bi bi-cpu"></i> PC Builder</h5>
                <p class="small">Build your custom PC with ease.</p>
            </div>
            <div class="col-md-6 text-md-end">
                <ul class="list-inline mb-0">
                    <li class="list-inline-item"><a href="index.html" class="text-white text-decoration-
none">Home</a></li>
                    <li class="list-inline-item"><a href="builder.html" class="text-white text-decoration-
none">Builder</a></li>
                    <li class="list-inline-item"><a href="suggest.html" class="text-white text-decoration-none">Suggest
Component</a></li>
                    <li class="list-inline-item"><a href="author.html" class="text-white text-decoration-none">About
Author</a></li>
                    <li class="list-inline-item"><a href="assets/other/doc.pdf" target="_blank" class="text-white text-
decoration-none">Documentation</a></li>
                </ul>
                <p class="small" id="footer-text">&copy; 2026 PC Builder. All rights reserved.</p>
            </div>
        </div>
    </div>

```

```

    </div>
  </footer>

  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
  <script src="assets/js/main.js"></script>
  <script src="assets/js/suggest.js"></script>
</body>

</html>

```

4.1.2. JS

main.js:

```

let allComponents = [];
let filteredComponents = [];
let buildComponents = [];
let favoriteComponentIds = [];
let savedBuilds = [];

let currentPage = 1;
const itemsPerPage = 6;

const LOCAL_STORAGE_KEY = 'pc-builder-build';
const FAVORITE_STORAGE_KEY = 'pc-builder-favorites';
const SAVED_BUILDS_STORAGE_KEY = 'pc-builder-saved-builds';
const COMPONENTS_URL = 'assets/data/components.json';

// ===== INITIALIZATION =====
document.addEventListener('DOMContentLoaded', function () {
  helpers.fixAria();
  helpers.footerText();
  setupEventListeners();

  if (document.getElementById('components-holder')) {
    loadComponents();
  }
});

// ===== DATA LOADING =====
function loadComponents() {
  fetch(COMPONENTS_URL)
    .then(response => {
      if (!response.ok) {
        throw new Error('Failed to load components: ${response.status}');
      }
    })
}

```

```

        return response.json();
    })
    .then(data => {
        allComponents = data;
        filteredComponents = data;
        loadSavedState();
        applyCategoryFromUrl();
        performFiltering();
        updateBuildDisplay();
    })
    .catch(error => {
        console.error(error);
        const componentsHolder = document.getElementById('components-holder');
        if (componentsHolder) {
            componentsHolder.innerHTML = '<div class="col-12"><div class="alert alert-danger">Components could
not be loaded.</div></div>';
        }
    });
}

// ===== EVENT LISTENERS =====
function setupEventListeners() {
    document.querySelectorAll('#search-desktop, #search-mobile').forEach(input => {
        input.addEventListener('input', performFiltering);
    });

    document.querySelectorAll('.form-check-input[name="category"]').forEach(checkbox => {
        checkbox.addEventListener('change', performFiltering);
    });

    document.querySelectorAll('#sort-desktop, #sort-mobile').forEach(select => {
        select.addEventListener('change', performFiltering);
    });

    document.querySelectorAll('.build-name-input').forEach(input => {
        input.addEventListener('input', function () {
            syncBuildNameInputs(this.value);
            updateBuildNameStatus(this.value);
        });
    });

    const componentModal = document.getElementById('componentModal');
    if (componentModal) {
        componentModal.addEventListener('show.bs.modal', function (event) {
            const componentId = event.relatedTarget?.getAttribute('data-component-id');

```

```

        showComponentDetails(componentId);
    });
}

document.addEventListener('click', function (event) {
    const addButton = event.target.closest('.add-to-build-btn');
    if (addButton) {
        const component = findComponentById(addButton.dataset.componentId);
        if (component) {
            toggleBuildComponent(component);
        }
        return;
    }

    const removeButton = event.target.closest('.remove-from-build-btn');
    if (removeButton) {
        removeFromBuild(removeButton.dataset.componentId);
        return;
    }

    const favoriteButton = event.target.closest('.favorite-icon');
    if (favoriteButton) {
        const componentId = favoriteButton.dataset.componentId;
        if (componentId) {
            toggleFavorite(componentId);
        }
        return;
    }

    const favoriteAddButton = event.target.closest('.add-favorite-to-build-btn');
    if (favoriteAddButton) {
        const component = findComponentById(favoriteAddButton.dataset.componentId);
        if (component) {
            addComponentToBuild(component);
        }
        return;
    }

    const favoriteRemoveButton = event.target.closest('.remove-favorite-btn');
    if (favoriteRemoveButton) {
        toggleFavorite(favoriteRemoveButton.dataset.componentId);
        return;
    }

    const saveBuildButton = event.target.closest('.save-build-btn');

```



```

    if (saveBuildButton) {
        saveCurrentBuild();
        return;
    }

    const loadSavedBuildButton = event.target.closest('.load-saved-build-btn');
    if (loadSavedBuildButton) {
        loadSavedBuild(loadSavedBuildButton.dataset.buildId);
        return;
    }

    const deleteSavedBuildButton = event.target.closest('.delete-saved-build-btn');
    if (deleteSavedBuildButton) {
        deleteSavedBuild(deleteSavedBuildButton.dataset.buildId);
        return;
    }

    const modalBuildButton = event.target.closest('#modal-build-button');
    if (modalBuildButton) {
        const component = findComponentById(modalBuildButton.dataset.componentId);
        if (component) {
            toggleBuildComponent(component);
            populateModal(component);
        }
    }
});
}

```

// ===== FILTERING & SORTING =====

```

function applyCategoryFromUrl() {
    const category = new URLSearchParams(window.location.search).get('category');
    if (!category) return;

    document.querySelectorAll('.form-check-input[name="category"][value="$
    {CSS.escape(category)}"]').forEach(checkbox => {
        checkbox.checked = true;
    });
}

```

```

function performFiltering() {
    currentPage = 1;

    const searchValue = getCurrentSearchValue();
    const checkedCategories = getCheckedCategories();
    const sortValue = getCurrentSortValue();
}

```

```

filteredComponents = allComponents.filter(component => {
  const matchesSearch = !searchValue || component.name.toLowerCase().includes(searchValue);
  const matchesCategory = checkedCategories.length === 0 || checkedCategories.includes(component.category);
  return matchesSearch && matchesCategory;
});

filteredComponents = sortComponents(filteredComponents, sortValue);
renderWithPagination();
}

function getCurrentSearchValue() {
  const desktopSearch = document.getElementById('search-desktop')?.value.trim();
  const mobileSearch = document.getElementById('search-mobile')?.value.trim();
  return (desktopSearch || mobileSearch || "").toLowerCase();
}

function getCheckedCategories() {
  return [...document.querySelectorAll('.form-check-input[name="category"]:checked')]
    .map(checkbox => checkbox.value);
}

function getCurrentSortValue() {
  const desktopSort = document.getElementById('sort-desktop')?.value;
  const mobileSort = document.getElementById('sort-mobile')?.value;

  if (window.matchMedia('(max-width: 767.98px)').matches) {
    return mobileSort || desktopSort || 'price-low-high';
  }

  return desktopSort || mobileSort || 'price-low-high';
}

function sortComponents(components, sortBy) {
  const sortedComponents = [...components];

  switch (sortBy) {
    case 'price-low-high':
      return sortedComponents.sort((a, b) => a.price - b.price);
    case 'price-high-low':
      return sortedComponents.sort((a, b) => b.price - a.price);
    case 'a-z':
      return sortedComponents.sort((a, b) => a.name.localeCompare(b.name));
    case 'z-a':
      return sortedComponents.sort((a, b) => b.name.localeCompare(a.name));
  }
}

```

```

    case 'newest':
        return sortedComponents.sort((a, b) => new Date(b.release_date) - new Date(a.release_date));
    case 'oldest':
        return sortedComponents.sort((a, b) => new Date(a.release_date) - new Date(b.release_date));
    default:
        return sortedComponents;
}
}

// ===== COMPONENT RENDERING =====
function renderWithPagination() {
    const totalPages = Math.ceil(filteredComponents.length / itemsPerPage);

    if (currentPage > totalPages) currentPage = totalPages || 1;
    if (currentPage < 1) currentPage = 1;

    const startIndex = (currentPage - 1) * itemsPerPage;
    const currentPageComponents = filteredComponents.slice(startIndex, startIndex + itemsPerPage);

    renderComponents(currentPageComponents);
    renderPaginationControls(totalPages);
}

function renderComponents(components) {
    const componentsHolder = document.getElementById('components-holder');
    if (!componentsHolder) return;

    if (components.length === 0) {
        componentsHolder.innerHTML = '<div class="col-12"><div class="alert alert-secondary">No components
match your filters.</div></div>';
        return;
    }

    componentsHolder.innerHTML = components.map(component => {
        const isInBuild = buildComponents.some(comp => comp.id === component.id);
        const sameCategoryComponent = buildComponents.find(comp => comp.category === component.category);
        const willReplace = sameCategoryComponent && sameCategoryComponent.id !== component.id;
        const isFavorite = favoriteComponentIds.includes(component.id);
        const buttonClass = isInBuild
            ? 'btn btn-danger btn-sm flex-grow-1 fw-bold add-to-build-btn'
            : 'btn btn-outline-success btn-sm flex-grow-1 fw-bold add-to-build-btn';
        const buttonText = getBuildButtonText(component, isInBuild, willReplace);
        const favoriteClass = isFavorite ? 'fa-solid fa-heart favorite-icon active' : 'fa-regular fa-heart favorite-icon';

        return `

```

```

<div class="col-12 col-lg-6 col-xl-4">
  <div class="card component-card h-100 position-relative">
    <i class="{favoriteClass}" title="Toggle favorite" data-component-id="$
{helpers.escapeHTML(component.id)}"></i>
    <div class="card-img-wrapper">
      
    </div>
    <div class="card-body d-flex flex-column">
      <div class="mb-2 d-flex flex-wrap gap-1">
        <span class="badge bg-primary">${helpers.prettyText(component.category)}</span>
        <span class="badge bg-secondary">${helpers.prettyText(component.brand)}</span>
      </div>
      <h6 class="card-title fw-bold">${helpers.escapeHTML(component.name)}</h6>
      <h4 class="text-success mt-auto pt-3">${helpers.formatPrice(component.price)}</h4>
      <div class="d-flex gap-2 mt-3">
        <button class="{buttonClass}" data-component-id="{helpers.escapeHTML(component.id)}">
          ${buttonText}
        </button>
        <button class="btn btn-outline-primary btn-sm" data-bs-toggle="modal" data-bs-
target="#componentModal" data-component-id="{helpers.escapeHTML(component.id)}">
          <i class="fa-solid fa-circle-info me-1"></i> Details
        </button>
      </div>
    </div>
  </div>
</div>
</div>
</div>
`;
}).join("");
}

```

```

function getBuildButtonText(component, isInBuild, willReplace) {
  if (isInBuild) {
    return '<i class="fa-solid fa-trash me-1"></i> Remove';
  }

  if (willReplace) {
    return '<i class="fa-solid fa-arrows-rotate me-1"></i> Replace ${helpers.prettyCategory(component.category)}';
  }

  return '<i class="fa-solid fa-plus me-1"></i> Add';
}

```

```

function renderPaginationControls(totalPages) {

```

```

const paginationBars = document.querySelectorAll('.pagination');
if (paginationBars.length === 0) return;

if (totalPages <= 1) {
    paginationBars.forEach(pagination => {
        pagination.innerHTML = "";
    });
    return;
}

let paginationHTML = "";

paginationHTML += currentPage === 1
    ? '<li class="page-item disabled"><span class="page-link">Previous</span></li>'
    : '<li class="page-item"><a class="page-link" href="#" data-page="{currentPage - 1}">Previous</a></li>';

getPagesToShow(currentPage, totalPages).forEach(page => {
    if (page === '...') {
        paginationHTML += '<li class="page-item disabled"><span class="page-link">...</span></li>';
    } else if (page === currentPage) {
        paginationHTML += '<li class="page-item active"><span class="page-link">${page}</span></li>';
    } else {
        paginationHTML += '<li class="page-item"><a class="page-link" href="#" data-page="{page}">${page}
</a></li>';
    }
});

paginationHTML += currentPage === totalPages
    ? '<li class="page-item disabled"><span class="page-link">Next</span></li>'
    : '<li class="page-item"><a class="page-link" href="#" data-page="{currentPage + 1}">Next</a></li>';

paginationBars.forEach(pagination => {
    pagination.innerHTML = paginationHTML;
});

document.querySelectorAll('.pagination .page-link[data-page]').forEach(link => {
    link.addEventListener('click', function (event) {
        event.preventDefault();
        currentPage = Number(this.dataset.page);
        renderWithPagination();
        window.scrollTo({ top: 0, behavior: 'smooth' });
    });
});
}

```

```

function getPagesToShow(current, total) {
  if (total <= 7) {
    return Array.from({ length: total }, (_, index) => index + 1);
  }

  const pages = [1];

  if (current <= 4) {
    for (let page = 2; page <= Math.min(5, total - 1); page++) {
      pages.push(page);
    }
    pages.push('...', total);
  } else if (current >= total - 3) {
    pages.push('...');
    for (let page = Math.max(total - 4, 2); page <= total; page++) {
      pages.push(page);
    }
  } else {
    pages.push('...', current - 1, current, current + 1, '...', total);
  }

  return pages;
}

// ===== MODAL DETAILS =====
function showComponentDetails(componentId) {
  const component = findComponentById(componentId);
  if (!component) return;
  populateModal(component);
}

function populateModal(component) {
  const modalTitle = document.querySelector('#componentModal .modal-title');
  const modalDisplayTitle = document.querySelector('#modal-title');
  const modalImage = document.querySelector('#modal-image');
  const modalSpecs = document.querySelector('#modal-specs');
  const modalPrice = document.querySelector('#modal-price');
  const modalCategory = document.querySelector('#modal-category');
  const modalBrand = document.querySelector('#modal-brand');
  const modalBuildButton = document.querySelector('#modal-build-button');

  if (modalTitle) modalTitle.textContent = component.name;
  if (modalDisplayTitle) modalDisplayTitle.textContent = component.name;
  if (modalImage) {
    modalImage.src = helpers.normalizeImagePath(component.image_path);
  }
}

```

```

    modalImage.alt = component.name;
  }
  if (modalPrice) modalPrice.textContent = helpers.formatPrice(component.price);
  if (modalCategory) modalCategory.textContent = helpers.prettyText(component.category);
  if (modalBrand) modalBrand.textContent = helpers.prettyText(component.brand);
  if (modalSpecs) modalSpecs.innerHTML = formatComponentSpecsRecursive(component.specs || {});

  if (modalBuildButton) {
    const isInBuild = buildComponents.some(comp => comp.id === component.id);
    const sameCategoryComponent = buildComponents.find(comp => comp.category === component.category);
    const willReplace = sameCategoryComponent && sameCategoryComponent.id !== component.id;
    modalBuildButton.dataset.componentId = component.id;
    modalBuildButton.className = getModalBuildButtonClass(isInBuild, willReplace);
    modalBuildButton.innerHTML = getModalBuildButtonText(component, isInBuild, willReplace);
  }
}

function getModalBuildButtonClass(isInBuild, willReplace) {
  if (isInBuild) return 'btn btn-danger fw-bold';
  if (willReplace) return 'btn btn-warning fw-bold';
  return 'btn btn-success fw-bold';
}

function getModalBuildButtonText(component, isInBuild, willReplace) {
  if (isInBuild) {
    return '<i class="fa-solid fa-trash me-1"></i> Remove from Build';
  }

  if (willReplace) {
    return '<i class="fa-solid fa-arrows-rotate me-1"></i> Replace ${helpers.prettyCategory(component.category)}
in Build';
  }

  return '<i class="fa-solid fa-plus me-1"></i> Add to Build';
}

function formatComponentSpecsRecursive(specs, level = 0) {
  return Object.entries(specs).map(([key, value]) => {
    const prettyKey = helpers.prettyText(key);

    if (typeof value === 'object' && value !== null && !Array.isArray(value)) {
      return `
        <div class="mb-3 ${level > 0 ? 'ps-3' : ''}">
          <h6 class="text-uppercase fw-bold ${level === 0 ? 'text-primary' : 'text-secondary'}">${
            prettyKey}</h6>

```

```

        <div class="${level > 0 ? 'ps-3' : ''}">
            ${formatComponentSpecsRecursive(value, level + 1)}
        </div>
    </div>
    `;
}

const displayValue = Array.isArray(value) ? value.join(', ') : value;
return `
    <div class="d-flex justify-content-between gap-3 py-2 border-bottom ${level > 0 ? 'ps-3' : ''}">
        <span class="text-muted">${prettyKey}</span>
        <span class="fw-medium text-end">${helpers.escapeHTML(String(displayValue))}</span>
    </div>
    `;
}).join("");
}

// ===== BUILD STATE =====
function toggleBuildComponent(component) {
    const existingIndex = buildComponents.findIndex(comp => comp.id === component.id);

    if (existingIndex !== -1) {
        buildComponents.splice(existingIndex, 1);
    } else {
        addOrReplaceBuildComponent(component);
    }

    saveBuildState();
    updateBuildDisplay();
    updateAllButtonStates();
}

function addComponentToBuild(component) {
    addOrReplaceBuildComponent(component);
    saveBuildState();

    updateBuildDisplay();
    updateAllButtonStates();
}

function addOrReplaceBuildComponent(component) {
    buildComponents = buildComponents.filter(existingComponent => existingComponent.category !==
component.category);
    buildComponents.push(component);
}

```



```

function removeFromBuild(componentId) {
  buildComponents = buildComponents.filter(comp => comp.id !== componentId);
  saveBuildState();
  updateBuildDisplay();
  updateAllButtonStates();
}

function updateAllButtonStates() {
  document.querySelectorAll('.add-to-build-btn').forEach(button => {
    const component = findComponentById(button.dataset.componentId);
    if (!component) return;

    const isInBuild = buildComponents.some(comp => comp.id === component.id);
    const sameCategoryComponent = buildComponents.find(comp => comp.category === component.category);
    const willReplace = sameCategoryComponent && sameCategoryComponent.id !== component.id;
    button.className = isInBuild
      ? 'btn btn-danger btn-sm flex-grow-1 fw-bold add-to-build-btn'
      : 'btn btn-outline-success btn-sm flex-grow-1 fw-bold add-to-build-btn';
    button.innerHTML = getBuildButtonText(component, isInBuild, willReplace);
  });
}

// ===== SIDEBAR RENDERING =====
function updateBuildDisplay() {
  renderBuildList('desktop-build-list', 'mb-2');
  renderBuildList('mobile-build-list', 'mb-3');
  renderCompatibilityAlerts('desktop-compatibility-alerts');
  renderCompatibilityAlerts('mobile-compatibility-alerts');
  renderFavoritesList('desktop-favorites-list', 'mb-2');
  renderFavoritesList('mobile-favorites-list', 'mb-3');
  renderSavedBuildsList('desktop-saved-builds-list', 'mb-2');
  renderSavedBuildsList('mobile-saved-builds-list', 'mb-3');
  updateBuildTotals();
  updateFavoriteCounts();
  updateSavedBuildCounts();
}

function renderBuildList(containerId, marginClass) {
  const container = document.getElementById(containerId);
  if (!container) return;

  if (buildComponents.length === 0) {
    container.innerHTML = '<p class="small text-white-50 mb-0">No components selected.</p>';
    return;
  }

```

```

    }

    container.innerHTML = buildComponents.map(component => `
      <div class="card bg-dark border-secondary ${marginClass} build-component">
        <div class="card-body p-2 d-flex justify-content-between align-items-center gap-2">
          <div class="overflow-hidden">
            <div class="small fw-bold text-truncate" title="${helpers.escapeHTML(component.name)}">${
helpers.escapeHTML(component.name)}</div>
            <div class="text-success small">${helpers.formatPrice(component.price)}</div>
          </div>
          <button class="btn btn-sm text-danger px-1 remove-from-build-btn" data-component-id="${
helpers.escapeHTML(component.id)}" title="Remove">
            <i class="fa-solid fa-xmark"></i>
          </button>
        </div>
      </div>
    `).join("");
  }

  // ===== COMPATIBILITY =====

  function renderCompatibilityAlerts(containerId) {
    const container = document.getElementById(containerId);
    if (!container) return;

    const alerts = getCompatibilityAlerts();
    if (alerts.length === 0) {
      container.innerHTML = "";
      return;
    }

    container.innerHTML = alerts.map(alert => `
      <div class="alert alert-${alert.type} py-2 px-2 small mb-2">
        <i class="fa-solid fa-triangle-exclamation me-1"></i>${helpers.escapeHTML(alert.message)}
      </div>
    `).join("");
  }

  function getCompatibilityAlerts() {
    const alerts = [];
    const processor = buildComponents.find(component => component.category === 'processor');
    const motherboard = buildComponents.find(component => component.category === 'motherboard');
    const memory = buildComponents.find(component => component.category === 'memory');
    const storage = buildComponents.find(component => component.category === 'storage');
    const powerSupply = buildComponents.find(component => component.category === 'power_supply');
    const power = calculatePowerSummary();
  }

```

```

if (processor && motherboard) {
  const cpuSocket = processor.specs?.socket;
  const motherboardSocket = motherboard.specs?.socket;
  if (cpuSocket && motherboardSocket && cpuSocket !== motherboardSocket) {
    alerts.push({
      type: 'danger',
      message: `CPU socket ${cpuSocket} does not match motherboard socket ${motherboardSocket}.`
    });
  }
}

if (motherboard && memory) {
  const motherboardMemoryType = motherboard.specs?.memory_type;
  const memoryType = memory.specs?.type;
  if (motherboardMemoryType && memoryType && motherboardMemoryType !== memoryType) {
    alerts.push({
      type: 'danger',
      message: `Motherboard supports ${motherboardMemoryType}, but selected RAM is ${memoryType}.`
    });
  }
}

if (motherboard && storage) {
  const storageSupport = getStorageCompatibility(motherboard, storage);
  if (!storageSupport.compatible) {
    alerts.push({
      type: 'danger',
      message: storageSupport.message
    });
  }
}

if (powerSupply) {
  const psuWattage = helpers.extractPsuWattage(powerSupply);
  if (psuWattage > 0 && psuWattage < power.requiredPower) {
    alerts.push({
      type: 'warning',
      message: `Selected PSU capacity is ${psuWattage} W. Estimated draw is ${power.currentPower} W; recommended PSU is ${power.requiredPower} W.`
    });
  }
}

return alerts;

```

```
}
```

```
function getStorageCompatibility(motherboard, storage) {  
  const connectors = motherboard.specs?.storage_connectors || {};  
  const formFactor = String(storage.specs?.form_factor || "");  
  const storageInterface = String(storage.specs?.interface || "");  
  const isM2 = formFactor.toLowerCase().includes('m.2') || storageInterface.toLowerCase().includes('pcie');  
  const isSata = storageInterface.toLowerCase().includes('sata');
```

```
  if (isM2 && Number(connectors.m2 || 0) <= 0) {  
    return {  
      compatible: false,  
      message: `Selected storage needs M.2/PCIe, but motherboard has no M.2 slots.`  
    };  
  }  
}
```

```
  if (isSata && Number(connectors.sata || 0) <= 0) {  
    return {  
      compatible: false,  
      message: `Selected storage needs SATA, but motherboard has no SATA connectors.`  
    };  
  }  
}
```

```
  return { compatible: true, message: "" };  
}
```

```
function renderFavoritesList(containerId, marginClass) {  
  const container = document.getElementById(containerId);  
  if (!container) return;  
  
  const favoriteComponents = favoriteComponentIds  
    .map(componentId => findComponentById(componentId))  
    .filter(Boolean);  
  
  if (favoriteComponents.length === 0) {  
    container.innerHTML = '<p class="small text-white-50 mb-0">No favorite components.</p>';  
    return;  
  }  
}
```

```
  container.innerHTML = favoriteComponents.map(component => {  
    const isInBuild = buildComponents.some(comp => comp.id === component.id);  
    return `  
      <div class="card bg-dark border-secondary ${marginClass}">  
        <div class="card-body p-2">  
          <div class="d-flex justify-content-between align-items-start gap-2">
```

```

        <div class="overflow-hidden">
          <div class="small fw-bold text-truncate" title="{helpers.escapeHTML(component.name)}">${
helpers.escapeHTML(component.name)}</div>
          <div class="text-success small">${helpers.formatPrice(component.price)}</div>
        </div>
        <button class="btn btn-sm text-danger px-1 remove-favorite-btn" data-component-id="{
helpers.escapeHTML(component.id)}" title="Remove favorite">
          <i class="fa-solid fa-xmark"></i>
        </button>
      </div>
      <button class="btn btn-outline-success btn-sm w-100 mt-2 add-favorite-to-build-btn" data-component-
id="{helpers.escapeHTML(component.id)}" ${isInBuild ? 'disabled' : ''}>
        ${isInBuild ? '<i class="fa-solid fa-check me-1"></i> Added to Build' : '<i class="fa-solid fa-plus
me-1"></i> Add to Build'}
      </button>
    </div>
  </div>
`;
  }).join("");
}

```

```

function renderSavedBuildsList(containerId, marginClass) {
  const container = document.getElementById(containerId);
  if (!container) return;

  if (savedBuilds.length === 0) {
    container.innerHTML = '<p class="small text-white-50 mb-0">No saved builds.</p>';
    return;
  }

  container.innerHTML = savedBuilds.map(build => {
    const components = build.componentIds
      .map(componentId => findComponentById(componentId))
      .filter(Boolean);
    const totalPrice = components.reduce((sum, component) => sum + Number(component.price || 0), 0);

    return `
      <div class="card bg-dark border-secondary ${marginClass}">
        <div class="card-body p-2">
          <div class="small fw-bold text-truncate" title="{helpers.escapeHTML(build.name)}">${
helpers.escapeHTML(build.name)}</div>
          <div class="small text-white-50">${components.length} components</div>
          <div class="small text-success">${helpers.formatPrice(totalPrice)}</div>
          <div class="d-flex gap-2 mt-2">

```

```

        <button class="btn btn-outline-primary btn-sm flex-grow-1 load-saved-build-btn" data-build-id="$
{helpers.escapeHTML(build.id)}">
            <i class="fa-solid fa-rotate me-1"></i> Load
        </button>
        <button class="btn btn-outline-danger btn-sm delete-saved-build-btn" data-build-id="$
{helpers.escapeHTML(build.id)}" title="Delete saved build">
            <i class="fa-solid fa-trash"></i>
        </button>
    </div>
</div>
</div>
`
    `;
    }).join("");
}

```

```

function updateBuildTotals() {
    const totalPrice = buildComponents.reduce((sum, comp) => sum + Number(comp.price || 0), 0);
    const power = calculatePowerSummary();

    setText('desktop-total-price', helpers.formatPrice(totalPrice));
    setText('desktop-current-power', `${power.currentPower} W`);
    setText('desktop-required-power', `${power.requiredPower} W`);
    setText('mobile-total-price', helpers.formatPrice(totalPrice));
    setText('mobile-current-power', `${power.currentPower} W`);
    setText('mobile-required-power', `${power.requiredPower} W`);
}

```

```

function calculatePowerSummary() {
    const currentPower = buildComponents.reduce((sum, component) => {
        return sum + helpers.estimateComponentPower(component);
    }, 0);

    const requiredPower = currentPower === 0 ? 0 : Math.ceil((currentPower * 1.3) / 50) * 50;

    return {
        currentPower,
        requiredPower
    };
}

```

```

// ===== FAVORITES =====
function toggleFavorite(componentId) {
    if (favoriteComponentIds.includes(componentId)) {
        favoriteComponentIds = favoriteComponentIds.filter(id => id !== componentId);
    } else {

```

```

        favoriteComponentIds.push(componentId);
    }

    saveFavorites();
    renderWithPagination();
    updateBuildDisplay();
}

function updateFavoriteCounts() {
    setText('desktop-favorite-count', favoriteComponentIds.length);
    setText('mobile-favorite-count', favoriteComponentIds.length);
}

function updateSavedBuildCounts() {
    setText('desktop-saved-build-count', savedBuilds.length);
    setText('mobile-saved-build-count', savedBuilds.length);
}

// ===== STORAGE =====
function loadSavedState() {
    const savedBuildIds = readStorageArray(LOCAL_STORAGE_KEY);
    favoriteComponentIds = readStorageArray(FAVORITE_STORAGE_KEY);
    savedBuilds = readStorageArray(SAVED_BUILDS_STORAGE_KEY).filter(build => {
        return build && typeof build.id === 'string' && typeof build.name === 'string' &&
Array.isArray(build.componentIds);
    });
    buildComponents = savedBuildIds
        .map(componentId => findComponentById(componentId))
        .filter(Boolean);
}

function saveBuildState() {
    localStorage.setItem(LOCAL_STORAGE_KEY, JSON.stringify(buildComponents.map(component =>
component.id)));
}

function saveFavorites() {
    localStorage.setItem(FAVORITE_STORAGE_KEY, JSON.stringify(favoriteComponentIds));
}

function saveSavedBuilds() {
    localStorage.setItem(SAVED_BUILDS_STORAGE_KEY, JSON.stringify(savedBuilds));
}

// ===== SAVED BUILDS =====

```

```

function saveCurrentBuild() {
  const buildName = getBuildName();

  if (!buildName) {
    setSaveBuildStatus('Enter a build name first.', true);
    return;
  }

  if (buildComponents.length === 0) {
    setSaveBuildStatus('Add at least one component before saving.', true);
    return;
  }

  const existingBuild = savedBuilds.find(build => build.name.toLowerCase() === buildName.toLowerCase());
  const componentIds = buildComponents.map(component => component.id);

  if (existingBuild) {
    existingBuild.componentIds = componentIds;
    existingBuild.updatedAt = new Date().toISOString();
  } else {
    savedBuilds.push({
      id: `build-${Date.now()}`,
      name: buildName,
      componentIds,
      updatedAt: new Date().toISOString()
    });
  }

  saveSavedBuilds();
  updateBuildDisplay();
  setSaveBuildStatus(existingBuild ? 'Saved build updated.' : 'Build saved.');
```

```

}

function loadSavedBuild(buildId) {
  const savedBuild = savedBuilds.find(build => build.id === buildId);
  if (!savedBuild) return;

  buildComponents = savedBuild.componentIds
    .map(componentId => findComponentById(componentId))
    .filter(Boolean);

  saveBuildState();
  setBuildName(savedBuild.name);
  updateBuildDisplay();
  renderWithPagination();

```



```

    setSaveBuildStatus(`Loaded "${savedBuild.name}"`);
}

function deleteSavedBuild(buildId) {
    savedBuilds = savedBuilds.filter(build => build.id !== buildId);
    saveSavedBuilds();
    updateBuildDisplay();
}

function getBuildName() {
    const input = [...document.querySelectorAll('.build-name-input')]
        .find(element => element.value.trim());
    return input ? input.value.trim() : "";
}

function setBuildName(name) {
    syncBuildNameInputs(name);
}

function syncBuildNameInputs(name) {
    document.querySelectorAll('.build-name-input').forEach(input => {
        input.value = name;
    });
}

function updateBuildNameStatus(name) {
    const buildName = String(name || "").trim();
    if (!buildName) {
        setSaveBuildStatus("");
        return;
    }

    const existingBuild = savedBuilds.find(build => build.name.toLowerCase() === buildName.toLowerCase());
    setSaveBuildStatus(existingBuild ? 'A saved build with this name will be updated.' : "", false, 'warning');
}

function setSaveBuildStatus(message, isError = false, tone = 'success') {
    document.querySelectorAll('.save-build-status').forEach(status => {
        status.textContent = message;
        status.classList.toggle('text-danger', isError);
        status.classList.toggle('text-warning', !isError && tone === 'warning' && Boolean(message));
        status.classList.toggle('text-success', !isError && tone === 'success' && Boolean(message));
        status.classList.toggle('text-white-50', !isError && !message);
    });
}

```

```
// ===== UTILITIES =====

function readStorageArray(key) {
  try {
    const value = JSON.parse(localStorage.getItem(key) || '[]');
    return Array.isArray(value) ? value : [];
  } catch {
    return [];
  }
}

function findComponentById(componentId) {
  return allComponents.find(component => component.id === componentId);
}

function setText(id, value) {
  const element = document.getElementById(id);
  if (element) {
    element.textContent = value;
  }
}

function formatComponentSpecs(specs) {
  return formatComponentSpecsRecursive(specs);
}

const helpers = {
  fixAria: function () {
    function handleHideFocus() {
      if (document.activeElement) {
        document.activeElement.blur();
      }
    }

    document.querySelectorAll('.modal').forEach(modal => {
      modal.addEventListener('hide.bs.modal', handleHideFocus);
    });

    document.querySelectorAll('.offcanvas').forEach(offcanvas => {
      offcanvas.addEventListener('hide.bs.offcanvas', handleHideFocus);
    });
  },

  prettyText: function (text) {
    return String(text || "")
  }
}
```

```
        .replace(/_/g, ' ')
        .replace(/\b\w/g, letter => letter.toUpperCase());
    },
```

```
prettyCategory: function (category) {
    const labels = {
        processor: 'CPU',
        graphics_card: 'GPU',
        power_supply: 'PSU'
    };

    return labels[category] || this.prettyText(category);
},
```

```
formatPrice: function (price) {
    return `$$${Number(price || 0).toFixed(2)}`;
},
```

```
normalizeImagePath: function (path) {
    return String(path || "").replace(/^\//, "");
},
```

```
extractWattage: function (component) {
    const tdp = component?.specs?.tdp;
    if (!tdp) return 0;

    const wattage = parseInt(String(tdp).replace('W', ''), 10);
    return Number.isNaN(wattage) ? 0 : wattage;
},
```

```
extractPsuWattage: function (component) {
    const wattage = component?.specs?.wattage;
    if (!wattage) return 0;

    const parsedWattage = parseInt(String(wattage).replace('W', ''), 10);
    return Number.isNaN(parsedWattage) ? 0 : parsedWattage;
},
```

```
estimateComponentPower: function (component) {
    const tdp = this.extractWattage(component);
    if (tdp > 0) return tdp;

    const estimatesByCategory = {
        motherboard: 50,
        memory: 8,
```

```

        storage: 8,
        monitor: 0,
        mouse: 2,
        keyboard: 3,
        headset: 3,
        power_supply: 0
    };

    return estimatesByCategory[component.category] || 0;
},

escapeHTML: function (value) {
    return String(value)
        .replace(/&/g, '&amp;')
        .replace(/</g, '&lt;')
        .replace(/>/g, '&gt;')
        .replace(/"/g, '&quot;')
        .replace(/'/g, '&#039;');
},

footerText: function () {
    const footer = document.getElementById('footer-text');
    if (footer) {
        footer.innerHTML = `&copy; ${new Date().getFullYear()} PC Builder. All rights reserved.`;
    }
}
};

```

suggest.js

```

document.addEventListener('DOMContentLoaded', function () {
    const form = document.getElementById('suggestForm');
    if (!form) return;

    const successMessage = document.getElementById('successMessage');
    const formMessage = document.getElementById('formMessage');
    const newSuggestionButton = document.getElementById('newSuggestion');

    const fields = {
        category: document.getElementById('componentCategory'),
        name: document.getElementById('componentName'),
        brand: document.getElementById('componentBrand'),
        price: document.getElementById('componentPrice'),
        releaseDate: document.getElementById('releaseDate'),
        specs: document.getElementById('componentSpecs'),
        email: document.getElementById('userEmail')
    }

```

```

};

const validators = {
  category: {
    test: value => Boolean(value),
    message: 'Please select a component category.'
  },
  name: {
    test: value => /^[A-Za-z0-9\s\-\.\()]{3,80}$/.test(value),
    message: 'Component name must be 3-80 characters and use only letters, numbers, spaces, +, /, dots,
hyphens, or parentheses.'
  },
  brand: {
    test: value => value === '' || /^[A-Za-z0-9\s\-\.&]{2,40}$/.test(value),
    message: 'Brand must be 2-40 characters and use only letters, numbers, spaces, dots, hyphens, or &.'
  },
  price: {
    test: value => value === '' || (/^\d+(\.\d{1,2})?$/).test(value) && Number(value) >= 0.01 && Number(value)
<= 9999.99),
    message: 'Price must be between 0.01 and 9999.99, with up to 2 decimal places.'
  },
  releaseDate: {
    test: value => value === '' || isValidReleaseDate(value),
    message: 'Release date cannot be in the future.'
  },
  specs: {
    test: value => value === '' || /^[A-Za-z0-9\s,.;:~\-\_\/()x"]{10,500}$/.test(value),
    message: 'Specifications must be 10-500 characters and use common specification symbols only.'
  },
  email: {
    test: value => /^[^s@]+\@[^s@]+\.[^s@]{2,}$/.test(value),
    message: 'Please enter a valid email address.'
  }
};

Object.entries(fields).forEach(([key, field]) => {
  field.addEventListener('input', function () {
    validateField(key);
    hideFormMessage();
    successMessage.classList.add('d-none');
  });

  field.addEventListener('change', function () {
    validateField(key);
    hideFormMessage();
  });
});

```

```
        successMessage.classList.add('d-none');
    });
});
```

```
form.addEventListener('submit', function (event) {
    event.preventDefault();
```

```
    const result = validateForm();
    if (!result.isValid) {
        showFormMessage(result.errors);
        successMessage.classList.add('d-none');
        return;
    }
```

```
    hideFormMessage();
    form.classList.add('d-none');
    successMessage.classList.remove('d-none');
});
```

```
form.addEventListener('reset', function () {
    setTimeout(function () {
        clearValidation();
        hideFormMessage();
        successMessage.classList.add('d-none');
    }, 0);
});
```

```
if (newSuggestionButton) {
    newSuggestionButton.addEventListener('click', function () {
        form.reset();
        clearValidation();
        hideFormMessage();
        successMessage.classList.add('d-none');
        form.classList.remove('d-none');
        fields.category.focus();
    });
}
```

```
function validateForm() {
    const errors = [];

    Object.keys(fields).forEach(key => {
        const error = validateField(key);
        if (error) {
            errors.push(error);
        }
    });
}
```

```

    }
  });

  return {
    isValid: errors.length === 0,
    errors
  };
}

function validateField(key) {
  const field = fields[key];
  const validator = validators[key];
  const value = field.value.trim();
  const isValid = validator.test(value);

  field.classList.toggle('is-invalid', !isValid);
  field.classList.toggle('is-valid', isValid && value !== "");

  return isValid ? "" : validator.message;
}

function showFormMessage(errors) {
  formMessage.className = 'alert alert-danger';
  formMessage.innerHTML = `
    <h5 class="alert-heading mb-2"><i class="bi bi-exclamation-triangle"></i> Please fix the form</h5>
    <ul class="mb-0">
      ${errors.map(error => `<li>${escapeHTML(error)}</li>`).join("")}
    </ul>
  `;
}

function hideFormMessage() {
  formMessage.className = 'alert d-none';
  formMessage.innerHTML = "";
}

function clearValidation() {
  Object.values(fields).forEach(field => {
    field.classList.remove('is-valid', 'is-invalid');
  });
}

function isValidReleaseDate(value) {
  const selectedDate = new Date(`${value}T00:00:00`);
  const today = new Date();

```

```

    today.setHours(0, 0, 0, 0);
    return !Number.isNaN(selectedDate.getTime()) && selectedDate <= today;
}

function escapeHTML(value) {
    return String(value)
        .replace(/&/g, '&amp;')
        .replace(/</g, '&lt;')
        .replace(/>/g, '&gt;')
        .replace(/"/g, '&quot;')
        .replace(/'/g, '&#039;');
}
});

```

4.1.3. JSON

components.json (SUMIRAN) [Jedna komponenta po kategoriji jer sam fajl ima 2000 linija koda]

```

[
  {
    "id": "cpu_1",
    "name": "AMD Ryzen 5 5600X 3.7 GHz 6-Core Processor",
    "specs": {
      "cores": 6,
      "threads": 12,
      "base_clock": "3.7 GHz",
      "boost_clock": "4.6 GHz",
      "tdp": "65W",
      "socket": "AM4",
      "cache": {
        "l1": "384 KB",
        "l2": "3 MB",
        "l3": "32 MB"
      }
    },
    "price": 159.99,
    "image_path": "assets/images/cpu_1.png",
    "category": "processor",
    "brand": "AMD",
    "release_date": "2020-11-05",
    "max_temperature": "95°C",
    "integrated_graphics": false
  },
  {

```



```
"id": "gpu_1",
"name": "NVIDIA GeForce RTX 3060 Ti 8 GB",
"specs": {
  "cuda_cores": 4864,
  "base_clock": "1410 MHz",
  "boost_clock": "1665 MHz",
  "memory_size": "8 GB",
  "memory_type": "GDDR6",
  "memory_bandwidth": "448 GB/s",
  "tdp": "200W"
},
"price": 399.99,
"image_path": "assets/images/gpu_1.png",
"category": "graphics_card",
"brand": "NVIDIA",
"release_date": "2020-12-02"
},
{
  "id": "motherboard_1",
  "name": "ASUS ROG Strix B550-F Gaming WiFi II",
  "specs": {
    "socket": "AM4",
    "chipset": "B550",
    "form_factor": "ATX",
    "memory_slots": 4,
    "max_memory": "128 GB",
    "memory_type": "DDR4",
    "pci_express_slots": {
      "x16": 2,
      "x1": 3
    },
    "storage_connectors": {
      "sata": 6,
      "m2": 2
    },
    "networking": {
      "ethernet_speed": "2.5 Gbps",
      "wifi_standard": "Wi-Fi 6E"
    }
  },
  "price": 189.99,
  "image_path": "assets/images/mb_asus_b550f.png",
  "category": "motherboard",
  "brand": "ASUS",
  "release_date": "2021-10-01"
```

```
},
{
  "id": "ram_1",
  "name": "Corsair Vengeance LPX 16 GB (2 x 8 GB)",
  "specs": {
    "capacity": "16 GB",
    "modules": "2 x 8 GB",
    "type": "DDR4",
    "speed": "3200 MHz",
    "cas_latency": 16,
    "voltage": "1.35V",
    "color": "Black",
    "rgb": false
  },
  "price": 44.99,
  "image_path": "assets/images/ram_corsair_lpx.png",
  "category": "memory",
  "brand": "Corsair",
  "release_date": "2015-08-01"
},
{
  "id": "psu_1",
  "name": "Corsair RM750x 750W",
  "specs": {
    "wattage": "750W",
    "efficiency_rating": "80+ Gold",
    "modularity": "Fully Modular",
    "form_factor": "ATX"
  },
  "price": 129.99,
  "image_path": "assets/images/psu_corsair_rm750x.png",
  "category": "power_supply",
  "brand": "Corsair",
  "release_date": "2021-04-15"
},
{
  "id": "storage_1",
  "name": "Samsung 970 Evo Plus 1 TB",
  "specs": {
    "capacity": "1 TB",
    "type": "NVMe SSD",
    "form_factor": "M.2-2280",
    "interface": "PCIe 3.0 x4",
    "sequential_read": "3500 MB/s",
    "sequential_write": "3300 MB/s"
  }
}
```

```
    },
    "price": 69.99,
    "image_path": "assets/images/storage_samsung_970evo.png",
    "category": "storage",
    "brand": "Samsung",
    "release_date": "2019-01-22"
  },
  {
    "id": "monitor_1",
    "name": "Dell S2721DGF",
    "specs": {
      "size": "27 inches",
      "resolution": "2560x1440",
      "refresh_rate": "165 Hz",
      "panel_type": "IPS",
      "response_time": "1 ms",
      "color_gamut": "98% DCI-P3",
      "connectivity": {
        "hdmi": 2,
        "displayport": 1,
        "usb_ports": {
          "usb_3_0": 4
        }
      }
    }
  },
  "price": 329.99,
  "image_path": "assets/images/monitor_dell_s2721dgf.png",
  "category": "monitor",
  "brand": "Dell",
  "release_date": "2020-07-15"
},
{
  "id": "mouse_1",
  "name": "Logitech G Pro X Superlight",
  "specs": {
    "type": "Wireless Gaming Mouse",
    "sensor": "HERO 25K",
    "dpi": "100 - 25,600",
    "buttons": 5,
    "battery_life": "70 hours",
    "weight": "63 grams",
    "connectivity": [
      "Wireless"
    ]
  }
},
```

```
"price": 149.99,
"image_path": "assets/images/mouse_logitech_superlight.png",
"category": "mouse",
"brand": "Logitech",
"release_date": "2020-12-03"
},
{
  "id": "keyboard_1",
  "name": "Razer BlackWidow V3",
  "specs": {
    "type": "Mechanical Gaming Keyboard",
    "switch_type": "Razer Green (Clicky)",
    "key_rollover": "N-Key Rollover",
    "backlighting": "RGB Chroma",
    "connectivity": [
      "Wired"
    ],
    "additional_features": [
      "Dedicated Media Controls",
      "Wrist Rest"
    ]
  },
  "price": 139.99,
  "image_path": "assets/images/kb_razer_blackwidow_v3.png",
  "category": "keyboard",
  "brand": "Razer",
  "release_date": "2020-08-18"
},
{
  "id": "headset_1",
  "name": "SteelSeries Arctis 7",
  "specs": {
    "type": "Wireless Gaming Headset",
    "driver_size": "40 mm",
    "frequency_response": "20 Hz - 20 kHz",
    "microphone": {
      "type": "Retractable Boom Mic",
      "noise_cancellation": true,
      "frequency_response": "100 Hz - 10 kHz"
    },
    "battery_life": "24 hours",
    "connectivity": [
      "Wireless",
      "Wired"
    ]
  }
}
```

```
    },  
    "price": 149.99,  
    "image_path": "assets/images/hs_steelseries_arctis7.png",  
    "category": "headset",  
    "brand": "SteelSeries",  
    "release_date": "2019-02-12"  
  }  
]
```